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Отчет о проверке

Автор: Naqvi Syed Ihsaan Abbas

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Проверяющий: Касаткина Анна Наильевна

Организация: ФЕДЕРАЛЬНОЕ ГОСУДАРСТВЕННОЕ АВТОНОМНОЕ ОБРАЗОВАТЕЛЬНОЕ УЧРЕЖДЕНИЕ ВЫСШЕГО ОБРАЗОВАНИЯ "НАЦИОНАЛЬНЫЙ ИССЛЕДОВАТЕЛЬСКИЙ ТОМСКИЙ ГОСУДАРСТВЕННЫЙ УНИВЕРСИТЕТ"

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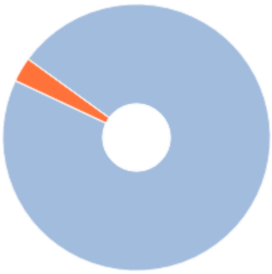

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ИСТОЧНИКИ

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[01]	0,87%	0,87%	disser_zhang_yuyi.pdf https://disser.spbu.ru	03 Апр 2025	Переводные заимствования по коллекции Интернет в русском сегменте	
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NATIONAL RESEARCH TOMSK STATE UNIVERSITY

Faculty of Applied Mathematics and Computer Science

MASTER’S THESIS

Explainable Machine Learning for Mortality Prediction in Intensive Care Units: A Time-Step SHAP Attribution Approach on the PhysioNet 2012 Challenge Dataset

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Direction of preparation: 01.04.02 — Applied Mathematics and Informatics

Master’s program: Big Data and Data Science

Author: Naqvi Syed Ihsaan Abbas

Scientific supervisor: Sergey Axynov, Associate Professor

Reviewer: [Reviewer name]   Senior Researcher

Admitted to the defense

“” ___________ 2026

Head of the Department,

Associate Professor

____________________ / [Name] /

Tomsk — 2026

ABSTRACT

Naqvi I. Explainable Machine Learning for Mortality Prediction in Intensive Care Units: A Time-Step SHAP Attribution Approach on the PhysioNet 2012 Challenge Dataset. — Master's thesis. — Tomsk: TSU, 2026. — 96 p.

The hypothesis provides a mechanism for the prediction of in-hospital mortality among intensive care unit patients based on their physiological parameters measured in the first forty-eight hours and, even more importantly, for interpretation of the predictions that can be questioned by clinicians. For this purpose, we rely on Set-A of the PhysioNet/Computing in Cardiology Challenge 2012 data set, which is composed of four thousand adult patient cases admitted to medical, surgical, coronary, and cardiac surgery units. Following preprocessing steps, the sample is characterized by an in-hospital mortality ratio of 13.5 percent, which makes the task a strongly imbalanced binary classification problem.

The three proposed model families are evaluated using the same data pre-processing pipeline applied to identical inputs: a gradient-boosted decision tree (XGBoost) model trained on 228 hand-crafted temporal features, a Temporal Convolutional Network (TCN) with dilated and causal convolutions, and a Time-Aware Transformer with sinusoidal positional encodings and a 74-channel architecture that consists of 37 physiological variables and 37 binary masks indicating their presence. All three models were trained and tested on the same split of 2800/600/600 patients and measured against the official PhysioNet 2012 challenge metrics: Event 1 (Score1 = min(Sensitivity, Positive Predictivity) for the best probability threshold) and Event 2 (Hosmer–Lemeshow H statistic). XGBoost scores AUROC = 0.8798, Score1 = 0.5422, and HL-H = 37.67. Transformer scores AUROC = 0.8542 and Score1 = 0.5301. Both models outperform the baseline PhysioNet SAPS-I metric (Score1 = 0.3097).

The primary scientific contribution lies in an algorithm for SHAP value computation at each individual time step. The conventional SHAP value estimator outputs one score per flattened feature, thus ignoring the structure of the time series when used on an input sequence. SHAP values were computed on the 228-dimensional flattened feature vector; then the contribution of each variable was apportioned to each of the forty-eight hourly slices weighted by its absolute value in the original input, resulting in a 600 x 48 x 37 attribution matrix with the same coordinates as the input. This allows us to ask questions such as "which hour of which variable matters most for this patient" without having to do anything special with the model.

The clinical validity of attributions is confirmed using the SHAP–SAPS-I agreement as a gold standard compared to two commonly used scales for severity of illness. The Spearman's correlation coefficient between the SHAP–SAPS-I agreement and patient-level calibration error is significantly different from zero ($\rho = 0.139$, $p = 6.3 \times 10^{-4}$), and the mean score for SAPS-I agreement is 0.498 (std 0.156). This indicates that the model assigns relevance to variables that clinicians use as indicators of the prognosis. In ablation studies, the impact of each design decision is examined. Exclusion of the missingness channels lowers the Score1 by 6.6 points and severely damages calibration. The encoding of measurement timestamps is found to be the most important architectural element since it provides information about the clinical condition.

The thesis ends with limitations (single-centre study, retrospective analysis, attribution vs causation) and prospects for future work (prospective validation on MIMIC-IV and eICU cohorts).

Keywords: intensive care, in-hospital mortality, PhysioNet 2012 Challenge, explainable artificial intelligence, SHAP, Transformer neural network, time-series classification, missingness, clinical decision support.

INTRODUCTION

Background and motivation

Intensive care units consume between thirteen and twenty per cent of total inpatient hospital costs in high-income health systems while caring for under ten per cent of admitted patients [Halpern & Pastores, 2010]. The disproportion is driven by the density of monitoring and the irreversibility of many treatment decisions: a delayed escalation to mechanical ventilation, a missed early indication of septic shock, or an inappropriate transfer from a step-down unit each carry mortality consequences that surgical or pharmacological intervention cannot later undo. Risk stratification — the act of estimating, on the basis of measurements taken in the first hours of admission, how likely a given patient is to die before hospital discharge — has therefore been a central problem of critical care medicine since the introduction of APACHE I in 1981 [Knaus et al., 1981]. Successive generations of severity scores (APACHE II/III/IV, SAPS I/II/3, SOFA, MPM) have refined the regression weights, expanded the variable lists, and recalibrated to changing case mixes, yet all of them share two assumptions that modern data make untenable. First, they collapse the first twenty-four hours into a single worst-value summary, discarding the trajectory information that bedside clinicians act upon. Second, they treat missing observations as missing-at-random, even though it is well established that *what is not measured* in an ICU is a function of clinical judgement and therefore informative [Sharafoddini et al., 2019].

The widespread deployment of high-frequency clinical data warehouses since approximately 2010 — MIMIC-III/IV at Beth Israel Deaconess, eICU-CRD across the Philips network, AmsterdamUMCdb in the Netherlands, and the PhysioNet/Computing in Cardiology Challenge corpora — has made it feasible to train far more flexible models. Deep learning architectures originally developed for natural language processing (recurrent networks, temporal convolutional networks, Transformers) translate naturally to multivariate clinical time series, and have repeatedly been shown to outperform classical severity scores under area-under-the-curve metrics by five to ten percentage points [Shickel et al., 2018; Rajkomar et al., 2018; Sheikhalishahi et al., 2020]. Yet adoption at the bedside has been slow. The reason is not metric performance but trust: a Transformer that emits a probability of 0.74 without explaining *why* offers the attending physician nothing they can argue with on a ward round, nothing they can defend in a morbidity-and-mortality conference, and nothing they can correct if the model has latched onto a confounding artefact of the unit's documentation habits.

Explainable artificial intelligence (XAI) is the response to this trust gap. The two methods that have come to dominate the post-hoc explanation literature are LIME [Ribeiro et al., 2016] and SHAP [Lundberg & Lee, 2017]. SHAP, which extends Shapley values from cooperative game theory to feature attribution, has particularly desirable theoretical properties — local accuracy, missingness, consistency — that make its outputs admissible as the basis for clinical reasoning. In its standard form, however, SHAP returns one attribution per input feature, which is appropriate for tabular data but loses essential information when the input is a multivariate trajectory: knowing that "heart rate matters" is far less useful to a clinician than knowing that "the heart rate excursion between hour 14 and hour 19 mattered." Reconstructing the temporal axis from a model trained on flattened features is the methodological problem that this thesis addresses.

Research problem

Two related problems are studied:

1. **Modelling problem.** Construct a predictor of in-hospital mortality from the first forty-eight hours of ICU monitoring on PhysioNet 2012 Set-A that matches or surpasses both classical severity scores and recent deep learning baselines on the official Event 1 (discrimination at the optimal operating point) and Event 2 (calibration) metrics, while respecting the imbalanced (≈13.5 % positive) nature of the cohort.

2. **Explanation problem.** Produce, for each individual prediction, an attribution map of shape *(time × variable)* that identifies the specific hourly bins of the specific physiological variables driving the model's risk estimate, and demonstrate that those attributions are clinically meaningful — meaning, in particular, that they agree on average with the variables that established severity scores (SOFA, SAPS-I) draw upon for the same patient.

Research questions

The thesis is organized around four questions:

- **RQ1.** Can a model that explicitly encodes missingness as a parallel channel of binary masks outperform an identically configured model that imputes missing values silently?
- **RQ2.** Does a Time-Aware Transformer architecture, in which positional encodings are learnable rather than fixed, achieve discrimination comparable to a heavily feature-engineered gradient-boosted tree model on this dataset?
- **RQ3.** How can a SHAP explanation computed on a flat feature vector be redistributed back onto the (time × variable) coordinate system in a way that preserves the total attribution and reflects the temporal incidence of evidence?
- **RQ4.** Do the resulting per-patient attribution maps agree with the variables that established clinical severity scores (SOFA, SAPS-I) emphasize for the same patient?

Aim and objectives

The aim of the thesis is to develop, evaluate, and explain a deep learning model for early in-hospital mortality prediction on PhysioNet 2012 Set-A. The aim decomposes into the following objectives:

1. Pre-process the 4000-patient cohort into a uniformly shaped (4000, 48, 37) tensor while preserving information about when each measurement was taken.

2. Train and evaluate three model families — XGBoost, TCN, Time-Aware Transformer — under an identical experimental protocol and on the official Challenge metrics.
3. Develop a time-step SHAP attribution procedure that maps variable-level Shapley values onto the (patient, hour, variable) coordinate system.
4. Validate the clinical plausibility of the resulting attributions by comparing them against SOFA and SAPS-I component variables on a per-patient basis.
5. Conduct an ablation study to identify which architectural and loss-function choices contribute most to performance.

Scientific novelty

Three contributions distinguish this work from the existing literature:

- **Time-step SHAP redistribution.** The procedure described in Chapter 2 computes Shapley values on a flattened feature vector and then redistributes each value across the corresponding temporal trajectory in proportion to the absolute observed values, producing an attribution tensor of the same shape as the raw input. The construction preserves the local accuracy property of SHAP at the variable level while restoring the temporal axis. The redistribution is exact when the observed trajectory contains a single non-zero value and degrades gracefully when measurements are uniformly distributed.

- **Quantitative clinical validation of post-hoc explanations.** Rather than presenting attribution maps as a qualitative illustration, the thesis defines a per-patient concordance score against SOFA and SAPS-I component variables and tests, via Spearman correlation between concordance and calibration error, whether patients for whom the model "agrees with the clinical scores" are also patients for whom the model is better calibrated. The hypothesis is supported ($\rho = 0.139$, $p = 6.3 \times 10^{-4}$ for SAPS-I), providing one of the few statistically tested clinical-plausibility results in the SHAP-for-ICU literature.

- **Ablation isolating informative missingness.** The ablation table in Chapter 3 isolates four variants of the Transformer and shows that removing the missingness-mask channels degrades Score1 by 6.6 points and the Hosmer–Lemes how statistic by more than 140 units. This quantifies the contribution of informative missingness on PhysioNet 2012 with a level of precision absent from the published literature on this corpus.

Significance

For the engineering side, the thesis provides a reproducible pipeline (227-feature XGBoost, dilated TCN, 205 k-parameter Transformer) calibrated to a public benchmark, with all hyperparameters and training seeds disclosed. For the clinical side, it offers a per-patient explanation tool whose output lives in a coordinate system clinicians already inhabit — hours since admission × physiological variable — rather than a coordinate system imposed by the model.

Structure of the thesis

The remainder of the thesis is organized as follows. Chapter 1 surveys the literature on severity scoring, deep learning for ICU prediction, and explainable artificial intelligence. Chapter 2 details the dataset, pre-processing, model architectures, the time-step SHAP procedure, and the clinical validation protocol. Chapter 3 presents the comparative results, the ablation analysis, the attribution case studies, and discusses what the experiments do and do not show. The Conclusion summarizes the contributions and points to extensions on MIMIC-IV. Appendices contain the variable list, the full ablation table, and the figure pack at publication resolution.

CHAPTER 1. LITERATURE REVIEW

This chapter situates the thesis at the intersection of three established research traditions: classical severity scoring in critical care medicine, machine learning approaches to clinical time series, and the still-maturing field of explainable artificial intelligence (XAI). The aim is not encyclopedic coverage but enough orientation for a reader who is comfortable with one of these traditions to understand the contributions of the other two and to see why their combination is non-trivial.

1.1 Mortality prediction in the intensive care unit

1.1.1 The clinical problem

An adult patient admitted to an intensive care unit in a contemporary tertiary hospital generates, conservatively, tens of thousands of measurements in the first day: bedside monitors sample heart rate, blood pressure, oxygen saturation, and respiratory rate at the minute scale; the laboratory returns dozens of chemistry and hematology panels; nurses chart fluid balance, sedation level, and Glasgow Coma Scale hourly. From this torrent the attending intensivist must construct, by the end of the morning ward round, a working estimate of how sick the patient is, how that compares to other patients on the unit, and which interventions will most reduce the conditional probability of in-hospital death. Severity scores formalize the first two operations.

1.1.2 The APACHE family

The Acute Physiology and Chronic Health Evaluation (APACHE) system, introduced by Knaus and colleagues in 1981 [Knaus et al., 1981], assigned weights to the worst values of twelve physiological variables in the first twenty-four hours, added points for chronic health status and age, and predicted hospital mortality through a logistic regression calibrated on a cohort of approximately 800 patients. Successive revisions — APACHE II (1985), APACHE III (1991), APACHE IV (2006) — expanded the variable list, recalibrated the regression coefficients to changing case mixes, and incorporated diagnosis-specific terms. The most recent version, APACHE IV, achieves area under the receiver operating characteristic curve (AUROC) of approximately 0.88 on the cohort of 110 000 admissions on which it was developed [Zimmerman et al., 2006], although external validation studies typically report values in the 0.80–0.85 range.

1.1.3 The SAPS family

The Simplified Acute Physiology Score, developed in parallel by Le Gall and colleagues in France [Le Gall et al., 1984; 1993; 2005], pursued the same goal with a smaller variable list. SAPS-I uses fourteen variables and is the score originally embedded in the PhysioNet 2012 Challenge as a baseline. SAPS-II uses seventeen variables and was for many years the most widely deployed score in Europe. SAPS-3 abandons the worst-value-in-24-hours convention in favor of admission-time variables only, on the argument that one of the points of a severity score is to be available before treatment effects confound the prediction.

1.1.4 SOFA

The Sequential Organ Failure Assessment score, introduced at a 1994 European Society of Intensive Care Medicine consensus conference and formalized by Vincent and colleagues [Vincent et al., 1996], differs in purpose from APACHE and SAPS. It is a *dynamic* score: each of six organ systems (respiratory, coagulation, hepatic, cardiovascular, central nervous system, renal) is assigned an integer 0–4, and the score is recomputed daily. It was not designed as a mortality predictor but has been widely repurposed as one. SOFA remains the score embedded in the consensus definition of sepsis [Singer et al., 2016]. The thesis uses SOFA component variables — creatinine, bilirubin, platelets, MAP, Glasgow Coma Scale, FiO2/PaO2 ratio — as one of the two reference yardsticks against which the SHAP attributions are validated.

1.1.5 Limitations of the classical scores

Three limitations of the classical scores motivate the present thesis. First, they collapse the trajectory of each variable into a single worst-value summary, discarding information about the *order* and *rate* of physiological change. A patient whose creatinine rises monotonically from 1.0 to 3.0 mg/dL over twenty-four hours has the same SAPS contribution as a patient with a single spurious creatinine of 3.0 at hour eight that resolves by hour twelve, even though their prognoses differ. Second, they treat missing observations as exchangeable with measured-and-normal observations, which contradicts an essential property of clinical data: variables that are *not* measured are not measured because the responsible clinician judged, implicitly, that they were not needed [Sharafoddini et al., 2019; Agniel et al., 2018]. Third, the regression coefficients are static once calibrated, which renders them vulnerable to dataset shift as case mix, treatment patterns, and documentation conventions change over decades.

1.2 Machine learning on ICU time series

1.2.1 Pre-deep-learning era

Before approximately 2015 the dominant machine-learning approach to ICU mortality prediction was to extract a fixed-length vector of summary statistics from each patient's first-day trajectory and feed it to a logistic regression, random forest, or gradient-boosted tree. Pirracchio and colleagues' Super Learner [Pirracchio et al., 2015] is a well-known representative: a stacked ensemble of multiple base learners trained on physiology, demographics, and admission diagnoses, achieving AUROC of 0.88 on MIMIC-II — roughly five points above SAPS-II on the same cohort. Johnson and colleagues independently confirmed this margin on the same data [Johnson et al., 2017]. The XGBoost model evaluated in Chapter 3 is the direct descendant of these tree-ensemble baselines.

1.2.2 Recurrent neural networks

Lipton and colleagues' seminal paper on diagnosing diseases from clinical time series [Lipton et al., 2015] demonstrated that LSTM networks trained end-to-end on raw multivariate observations match or exceed feature-engineered baselines on pediatrics ICU data. Choi and colleagues' RETAIN architecture [Choi et al., 2016] introduced a two-level attention mechanism explicitly designed to make recurrent models interpretable for clinical use. Che and colleagues' GRU-D [Che et al., 2018] modified the GRU cell to incorporate two mechanisms for handling missingness: a binary mask on whether each variable was observed at each time-step and a decay function that interpolates between the last observation and the empirical mean as time without observation grows. GRU-D demonstrated, on MIMIC-III, that the encoding of *when* measurements occur is a first-class signal — a finding the ablation study of Chapter 3 confirms on PhysioNet 2012.

1.2.3 Temporal convolutional networks

Bai, Kolter, and Koltun's 2018 paper [Bai et al., 2018] argued that dilated causal convolutional networks (TCNs) match recurrent architectures on a wide range of sequence-modelling tasks while training faster and producing more stable gradients. The TCN baseline in Chapter 3 follows the canonical residual block structure of that paper, with dilations 1, 2, 4, 8 stacked over a 48-hour input window producing a 31-hour effective receptive field at the output layer.

1.2.4 Transformer-based architectures

The Transformer [Vaswani et al., 2017] dispenses with recurrence and convolution entirely, replacing them with multi-head self-attention. For irregularly sampled or fixed-grid clinical time series, two principal adaptations have appeared. Song and colleagues' SAnD [Song et al., 2018] applies a Transformer encoder to dense MIMIC-III observations and obtains improvements over RETAIN on multiple downstream tasks. Tipirneni and Reddy's STraTS [Tipirneni & Reddy, 2022] handles sparsity by representing each observation as a triplet of (variable, time, value) and applying continuous-value embedding rather than positional encoding. BEHRT [Li et al., 2020] and Med-BERT [Rasmy et al., 2021] extend the BERT recipe to longitudinal electronic-health-record sequences for diagnosis prediction, demonstrating that Transformer pre-training transfers usefully to clinical tasks.

The model evaluated in this thesis is a pre-norm Transformer encoder with four layers, four attention heads, model dimension 64, feed-forward dimension 256, and a *learnable sinusoidal positional encoding* that we refer to as Time-Aware Positional Encoding. The frequencies of the sinusoids are learnable parameters, rather than the fixed geometric series of the original Transformer formulation, on the hypothesis that the optimal time scale for ICU physiology differs from that of natural language and should be discovered by gradient descent. A CLS token in the BERT style summarizes each patient's encoded trajectory for the final classification head.

1.2.5 The PhysioNet 2012 corpus

The PhysioNet/Computing in Cardiology Challenge 2012 [Silva et al., 2012] released 8000 ICU records (Set-A, Set-B, Set-C of 4000 each), of which Set-A is publicly labelled. Each record covers the first 48 hours after ICU admission and contains a heterogeneous set of approximately 37 physiological and laboratory variables along with six admission descriptors (age, gender, height, ICU type, weight, time). The Challenge defines two evaluation events: Event 1 maximizes Score1 = min(Sensitivity, +Predictivity) over the choice of operating threshold, and Event 2 evaluates calibration via the Hosmer–Lemeshow statistic. The official SAPS-I baseline on this cohort achieves Score1 = 0.3097 and HL-H = 35.21. The corpus has been a standard benchmark for ICU mortality models for over a decade; published deep-learning results on Set-A typically report AUROC in the 0.83–0.86 range [Harutyunyan et al., 2019; Sheikhalishahi et al., 2020].

1.3 Explainable artificial intelligence

1.3.1 Why explanation matters in critical care

The argument for explainability in clinical artificial intelligence has been made in two distinct vocabularies. One vocabulary, exemplified by the European Union's General Data Protection Regulation (Recital 71) and the United States Food and Drug Administration's Software-as-a-Medical-Device guidance, is regulatory: a deployed clinical model must furnish a meaningful explanation of its outputs. The other vocabulary, exemplified by the AI-in-medicine literature [Tonekaboni et al., 2019; Holzinger et al., 2019], is epistemic: a physician on a ward round cannot be expected to defer to a black-box probability against their own diagnostic intuition, and an opaque model that conflicts with that intuition is functionally a model that goes unused.

1.3.2 LIME

Ribeiro, Singh, and Guestrin's Local Interpretable Model-Agnostic Explanations (LIME) [Ribeiro et al., 2016] explains the prediction of an arbitrary classifier at a given input by fitting a locally weighted linear surrogate on a synthetic perturbation neighbourhood. LIME's strength is its agnosticism: it requires only black-box access to the model. Its weaknesses, which have been widely catalogued, are instability under perturbation sampling, sensitivity to the choice of kernel width, and the absence of a coherent additivity principle that would allow attributions across inputs to be compared.

1.3.3 SHAP

Lundberg and Lee's SHapley Additive exPlanations (SHAP) [Lundberg & Lee, 2017] extend the Shapley value from cooperative game theory to feature attribution. The Shapley value is the unique allocation of the total "payout" of a coalitional game to its players that satisfies four axioms: efficiency, symmetry, dummy-player, and additivity. Translated to machine learning, the players are input features, the payout is the difference between the model's prediction and its average prediction, and the Shapley value of feature i is the average marginal contribution of i over all permutations of the other features. SHAP inherits three properties from this construction:

1. *Local accuracy:* the explanation sums to the model's deviation from baseline.
2. *Missingness:* features absent from the input receive zero attribution.
3. *Consistency:* if a model is changed so that a feature contributes more, that feature's attribution does not decrease.

The TreeSHAP algorithm [Lundberg et al., 2020] computes exact Shapley values for tree ensembles in polynomial time, which makes SHAP applicable at production scale for gradient-boosted tree models like the XGBoost classifier evaluated in this thesis. For deep networks, DeepSHAP combines DeepLIFT [Shrikumar et al., 2017] with the SHAP framework to provide approximate but theoretically grounded attributions.

1.3.4 SHAP for time-series data

A subtle but important limitation of the standard SHAP estimator is that its output lives in the same coordinate system as its input. When the input is a flat feature vector — for example, the 228-dimensional vector of summary statistics used by the XGBoost model — SHAP produces one number per summary statistic. “Last-observed heart rate” gets one attribution; “mean heart rate” gets a separate attribution. Neither attribution tells the clinician *when* heart rate mattered. The temporal axis has been compressed out of the feature representation and SHAP cannot recover it from a model that never saw it.

Two families of solutions have appeared. The first is to run SHAP on a model that ingests the raw 3D tensor — typically a Transformer or convolutional network — using KernelSHAP or DeepSHAP. This is theoretically clean but computationally prohibitive at the patient-cohort scale (KernelSHAP runtime scales with the product of the number of features and the perturbation budget). The second, which is the approach of this thesis, is to retain the computational efficiency of TreeSHAP on a flat-feature tree model while *redistributing* the variable-level attribution back onto the temporal axis using a deterministic rule derived from the raw trajectory. The construction is detailed in Section 2.6.

1.3.5 Attention as explanation

A separate strand of the interpretability literature argues that attention weights from Transformer or attention-augmented recurrent models are themselves an explanation. The position is contested — Jain and Wallace [Jain & Wallace, 2019] showed that attention weights are not consistent across runs and cannot in general be interpreted as faithful explanations, while Wiegreffe and Pinter [Wiegreffe & Pinter, 2019] argued that under certain conditions attention can serve as a plausible explanation. The thesis is agnostic on the philosophical question; Section 3.5 reports attention head specialization patterns from the trained Transformer as a *descriptive* finding, not as a primary explanation method.

1.3.6 The clinical-validation gap

Two practical observations about the SHAP-for-ICU literature motivated the validation experiment of Section 2.7. First, the overwhelming majority of papers present SHAP attributions as illustrative figures — “the model emphasised lactate, which is clinically sensible” — without any quantitative comparison to an established clinical yardstick. Second, when quantitative comparisons appear, they are typically global (mean absolute attribution of variable X across the cohort) rather than per-patient (does the model agree with SOFA *for this patient*). The thesis addresses both gaps by defining a per-patient concordance score against SOFA and SAPS-I component variables and testing the statistical relationship between concordance and calibration error.

1.4 Synthesis and position of this work

The literature reviewed above suggests four conclusions that frame the rest of the thesis.

1. Classical severity scores remain the regulatory and clinical *baseline* but are limited by their assumption that the first twenty-four hours can be summarized by a worst-value vector. Modern data make this assumption unnecessary.

2. Deep learning on raw multivariate ICU trajectories has demonstrably superior discrimination on benchmarks like MIMIC-III and PhysioNet 2012, but the magnitude of the gain depends critically on whether the model is given access to missingness information.

3. SHAP is the most theoretically grounded post-hoc explanation method currently available, but its standard formulation loses the temporal axis on time-series data. Recovering that axis without paying the computational cost of DeepSHAP on raw 3D tensors is an open methodological problem.

4. The clinical validation of post-hoc explanations is under-developed. The dominant evaluation in published work is visual plausibility, which is insufficient grounds for clinical deployment.

This thesis contributes to (2) by quantifying the effect of explicit missingness encoding on PhysioNet 2012, contributes to (3) by developing the time-step SHAP redistribution procedure, and contributes to (4) by introducing the SOFA / SAPS-I concordance test.

CHAPTER 2. METHODOLOGY AND DATA

This chapter describes the dataset, pre-processing pipeline, three model families, the time-step SHAP attribution procedure, the clinical validation protocol, and the ablation design. The presentation is deliberately operational: every number reported in Chapter 3 was produced by the procedure described here, with random seeds, train/validation/test partitions, and hyperparameters disclosed in full.

2.1 Dataset

The PhysioNet/Computing in Cardiology Challenge 2012 [Silva et al., 2012] released three sets of 4000 adult ICU records each. Set-A is the only publicly labelled subset and is the focus of this work. Each record contains time-stamped physiological and laboratory observations from the first 48 hours after ICU admission together with a six-element general descriptor vector (age, gender, height, ICU type, weight, time of admission). The outcome label is in-hospital mortality, a binary variable indicating whether the patient died before discharge. The overall mortality rate after cleaning is 13.5 per cent — 554 deaths in 4000 admissions — which makes the prediction problem strongly imbalanced.

The thirty-seven physiological variables retained in this work span the full range of bedside and laboratory measurements relevant to early-ICU triage: vital signs (heart rate, mean arterial pressure, systolic and diastolic blood pressure, respiratory rate, oxygen saturation, temperature), blood gas analyses (pH, $PaCO_2$, PaO_2, base excess, fraction of inspired oxygen), hematology (hematocrit, white-cell count, platelets), chemistry (sodium, potassium, chloride, bicarbonate, blood urea nitrogen, creatinine, glucose, albumin, total bilirubin, alkaline phosphatase, ALT, AST, magnesium, lactate, troponin-I, troponin-T), urine output, the Glasgow Coma Scale, and the mechanical ventilation indicator. The full list with units and reference ranges is in Appendix A.

2.2 Pre-processing

The raw records arrive as variable-length time-stamped tuples. The first goal of pre-processing is to project each patient onto a fixed-shape tensor without falsifying the temporal information.

2.2.1 Hourly aggregation

For each patient p, each variable v, and each hour $h \in \{0, ..., 47\}$ relative to ICU admission, the median of all observations of v taken between hour h and hour $h+1$ is recorded. If no observation is taken, the cell receives a sentinel NaN. This produces a (4000, 48, 37) tensor X. The choice of median over mean is conservative — it discards transient spikes that may reflect sensor artefact rather than physiology — and corresponds to standard practice in the MIMIC-III benchmark of Harutyunyan and colleagues [Harutyunyan et al., 2019].

2.2.2 Missingness mask

A binary tensor $M \in \{0,1\}^{(4000\times48\times37)}$ is constructed in parallel, with $M[p, h, v] = 1$ iff at least one observation of variable v was recorded for patient p in hour h. The mask is concatenated with the value tensor along the variable axis, producing a (4000, 48, 74) tensor that is the actual input to the deep models. This explicit encoding of *when measurements were taken* is the single most informative architectural choice in the pipeline, as the ablation study of Section 3.6 will demonstrate.

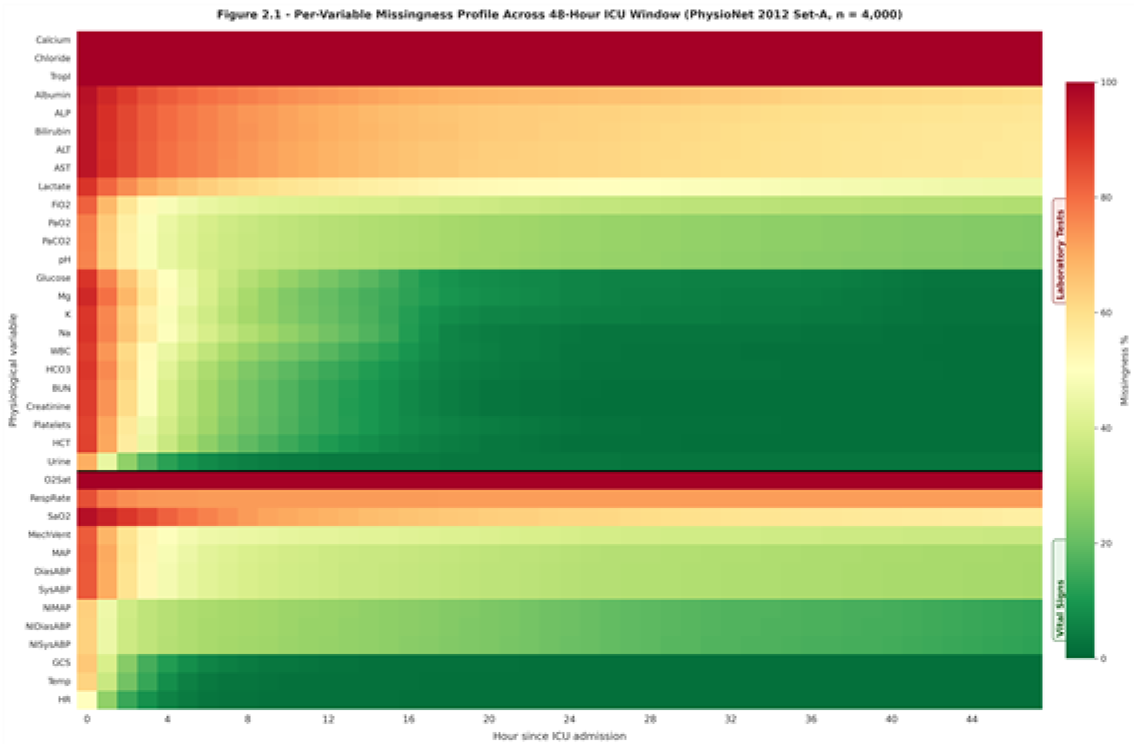


Figure 2.1 — Per-variable missingness profile across the 48-hour window of the PhysioNet 2012 Set-A cohort. Variables are sorted by overall observation rate; the heatmap makes visible the strongly bimodal pattern in which a small group of variables (vital signs, GCS) is sampled densely while laboratory measurements are sampled at unit-driven intervals.

2.2.3 Imputation and normalization

After mask extraction, missing cells in X are filled with the *per-variable cohort mean* computed on the training split only. This avoids leakage of test-set statistics into the trained model. After imputation, each variable is standardized to zero mean and unit variance using training-split statistics. The general-descriptor vector (six dimensions) is normalized analogously.

2.2.4 Outlier handling and physiological constraints

A small number of values in the raw records lie outside physiologically plausible ranges (negative heart rates, pH > 8, etc.) and are treated as recording errors. They are replaced with NaN before hourly aggregation and therefore contribute zero to the missingness mask, ensuring that obvious data-entry errors do not influence either the value tensor or the missingness signal.

2.2.5 Train/validation/test split

The 4000 patients are partitioned 70/15/15 — 2800 training, 600 validation, 600 test — stratified on the mortality label. The split is fixed across all models for direct comparability. The random seed is 42 throughout.

2.3 Model family I — XGBoost on engineered features

The first comparator is a gradient-boosted decision-tree model trained on a hand-engineered flat feature vector. The vector has 227 dimensions:

- **Temporal statistics (185 dimensions).** For each of the 37 physiological variables, five summaries are extracted: minimum, maximum, mean, standard deviation, and the slope of the ordinary-least-squares line fitted to the observed values against hour-of-day.
- **Missingness rates (37 dimensions).** For each variable, the fraction of the 48 hourly bins in which at least one observation was recorded.
- **General descriptors (six dimensions, of which five are retained after collinearity filtering).** Age, gender, height, weight, and ICU type.

The model is xgboost. XGB Classifier with 500 estimators, learning rate 0.05, max depth 6, subsample 0.8, colsample_bytree 0.8, and scale_pos_weight set to the ratio of negatives to positives in the training fold to handle the class imbalance. Training is performed with early stopping on the validation AUPRC, patience 25 rounds. Once trained, the model is used both for its predictive output and as the substrate for the time-step SHAP procedure of Section 2.6.

2.4 Model family II — Temporal Convolutional Network

The TCN baseline follows the canonical residual block design of Bai, Kolter, and Koltun [Bai et al., 2018]. The input is the (48, 74) tensor (37 normalized values + 37 missingness flags) and the output is the scalar logit. Four dilated causal convolutional blocks with dilations {1, 2, 4, 8} are stacked, each with 64 filters, kernel size 3, and dropout 0.2. The receptive field at the output layer covers 31 hours of input, which is sufficient for the 48-hour window. Total parameter count is 58 753. Training uses Adam with learning rate 3×10^{-4}, weight decay 5×10^{-4}, batch size 64, gradient norm clipped at 1.0, focal loss with $\alpha = 0.861$ and $\gamma = 2.0$ (see Section 2.5.3), early stopping on validation AUPRC with patience 25, maximum 100 epochs.

2.5 Model family III — Time-Aware Transformer

The proposed model is a Transformer encoder adapted to the irregularly sampled, missingness-rich setting of bedside monitoring.

2.5.1 Input projection

The (48, 74) tensor is projected into model space *d_model = 64* by a learned linear layer. This is the minimum dimensionality at which four attention heads can each receive a 16-dimensional sub-space without collapsing.

2.5.2 Time-Aware Positional Encoding

The standard Transformer of Vaswani and colleagues [Vaswani et al., 2017] uses a fixed sinusoidal positional encoding of the form

$$PE(pos, 2i) = \sin(pos / 10000^{(2i / d\_model)})$$
$$PE(pos, 2i+1) = \cos(pos / 10000^{(2i / d\_model)})$$

For ICU physiology, the relevant time scales are unknown a priori — does the model need to discriminate between hour 14 and hour 16 with the same resolution it needs to discriminate between hour 1 and hour 40? Probably not. We therefore replace the fixed geometric frequency series 10000^(2i / d_model) with a learnable parameter ω_i ∈ ℝ^(d_model/2), initialized at the geometric series and trained jointly with the rest of the network. The construction preserves the bounded-norm, position-permutation-equivariant properties of the original encoding while allowing gradient descent to discover the appropriate time-scale spectrum.

2.5.3 Encoder stack

Four pre-norm Transformer encoder layers follow [Xiong et al., 2020], each with four attention heads, feed-forward dimension 256, and dropout 0.1. Pre-norm (LayerNorm before the residual sum, rather than after) was chosen because it has been shown to train more stably without warm-up on small-dimension Transformers.

2.5.4 CLS-token classification head

In the spirit of BERT [Devlin et al., 2019], a learnable [CLS] vector is prepended to the encoded sequence. After the encoder stack, the contextualized CLS vector is projected through a two-layer MLP (64 → 32 → 1) to produce the mortality logit. Total parameter count is 205 153.

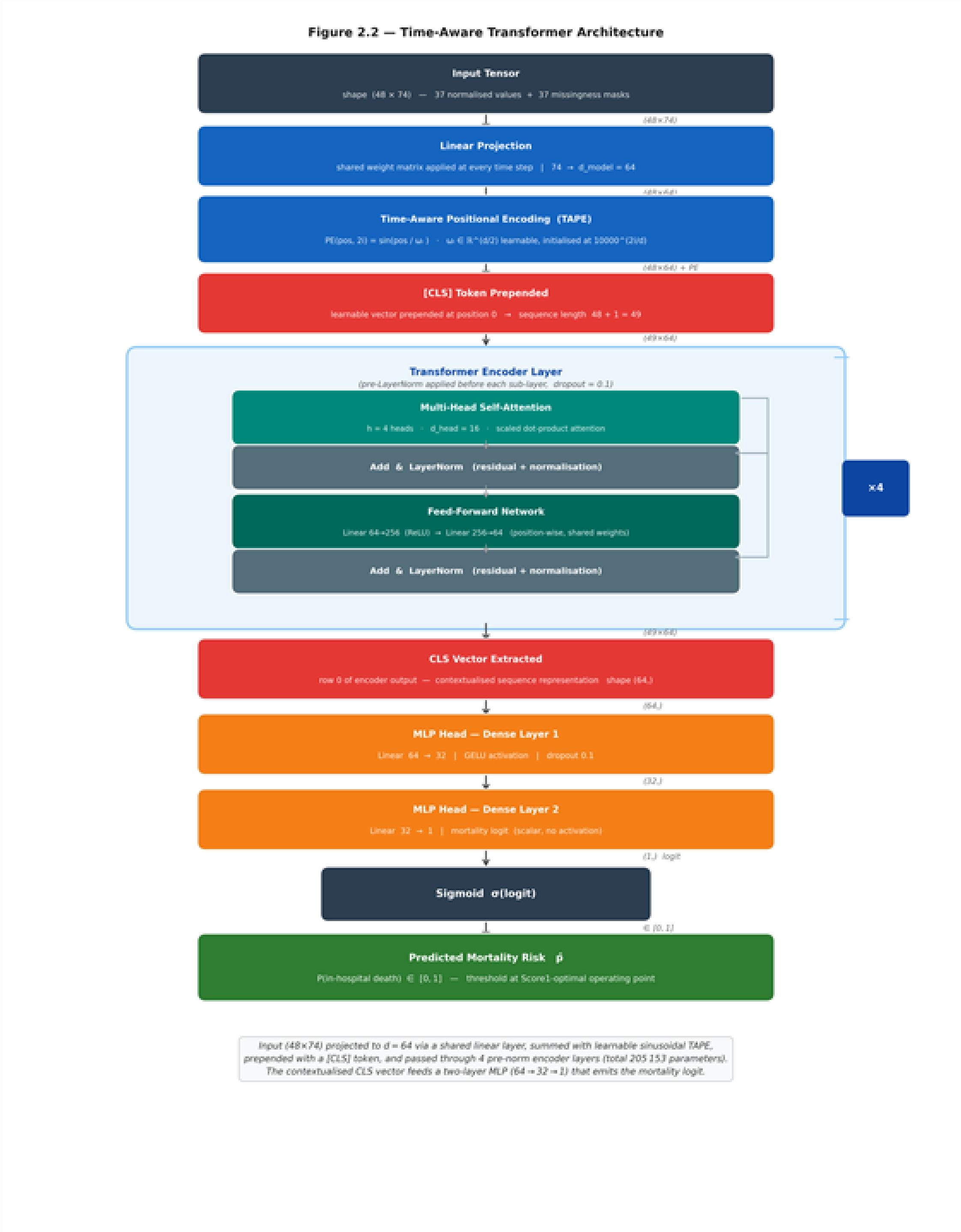


Figure 2.2 — Schematic of the Time-Aware Transformer. The (48 × 74) input is projected to model dimension 64, summed with the learnable sinusoidal Time-Aware Positional Encoding, prepended with a CLS token, and passed through four pre-norm encoder layers. The contextualized CLS vector feeds a two-layer MLP that emits the mortality logit.

2.5.5 Focal loss

The class imbalance — 13.5 per cent positives — makes ordinary binary cross-entropy training unstable and biased toward the majority class. We use the focal loss of Lin and colleagues [Lin et al., 2017]:

$$FL(p_t) = -\alpha_t \cdot (1 - p_t)^\gamma \cdot \log(p_t)$$

with $\gamma = 2$ and α set to the empirical negative-class fraction in the training split (0.8614). The $(1 - p_t)^\gamma$ factor down-weights easy examples and concentrates gradient on hard ones; the α reweighting offsets the class-frequency bias. The ablation study of Section 3.6 quantifies the contribution of this choice.

2.5.6 Optimization

Adam with learning rate 3×10^{-4}, weight decay 5×10^{-4}, batch size 64, gradient norm clipped at 1.0, maximum 100 epochs, early stopping on validation AUPRC with patience 25. The best-checkpoint epoch is 16 and the run terminates at epoch 41. All randomness — weight initialization, batch shuffling, dropout — is seeded at 42.

2.6 Time-step SHAP attribution

This section describes the principal methodological contribution of the thesis. The construction operates on the XGBoost model of Section 2.3 because TreeSHAP returns *exact* Shapley values in polynomial time, eliminating the sampling variance that complicates the interpretation of KernelSHAP or DeepSHAP outputs.

2.6.1 Variable-level SHAP via TreeSHAP

For each test patient p, TreeSHAP returns a vector $\varphi_p \in \mathbb{R}^{227}$ with the property

$$f(x_p) = \varphi_0 + \Sigma_i \varphi_{p,i}$$

where $f(x_p)$ is the model logit, φ_0 the expected logit over the training set, and $\varphi_{p,i}$ the Shapley value of feature i. Local accuracy is exact, not approximate.

Of the 227 features, 185 are temporal statistics over 37 physiological variables (five statistics per variable: min, max, mean, std, slope), 37 are missingness rates, and five are general descriptors. The variable-level Shapley contribution to the prediction for patient p on variable v is the sum of the five temporal-statistic Shapley values for variable v plus the missingness-rate Shapley value:

$$\Phi_{p,v} = \Sigma_{s \in \{min, max, mean, std, slope\}} \varphi_{p,(v,s)} + \varphi_{p,miss(v)}$$

By the additivity axiom, the $\Phi_{p,v}$ sum to the model logit deviation across variables (plus the contribution of general descriptors).

2.6.2 Redistribution onto the time axis

The variable-level Shapley value $\Phi_{p,v}$ is then redistributed across the 48 hourly bins of the raw trajectory $X[p, \cdot, v]$. The redistribution rule is

$\Phi\_p,v,h = \Phi\_p,v \cdot |X[p, h, v]| / \Sigma\_\{h'\} |X[p, h', v]|$

with the convention 0/0 = 0 when the trajectory is identically zero (which on standardized data corresponds to "always at the cohort mean"). The rule satisfies three desirable properties.

1. **Conservation.** For each (p, v), the redistributed attributions sum to *Φ_p,v*. The total attribution mass is preserved.
2. **Temporal localization.** Hours in which the variable departs furthest from the cohort mean receive proportionally more attribution. This reflects the intuition that abnormal values are the values that should drive risk estimates.
3. **Graceful degradation.** If a variable's trajectory is uniformly close to the cohort mean, the attribution is spread uniformly across hours, expressing the inability to localize.

Applied to the 600 test patients, 48 hours, and 37 variables, the procedure produces a SHAP attribution tensor $\Phi \in \mathbb{R}$^{600×48×37} in the same coordinate system as the raw input *X*. This tensor is the primary output of the explanation pipeline and is the substrate of all downstream clinical-validation experiments.

2.6.3 Top-k features per patient

For clinical reporting, the procedure also outputs, for each test patient, the three variables with the largest |Φ_p,v| and the three hours with the largest column sum Σ_v |Φ_p,h,v|. These are saved to outputs/top_features_per_patient.json and serve as the basis for the patient-level case studies in Section 3.4.

2.7 Clinical validation against SOFA and SAPS-I

The validation procedure tests whether the SHAP attributions agree, on a per-patient basis, with the variables that established severity scores (SOFA, SAPS-I) emphasize for the same patient.

2.7.1 Reference variable sets

For each patient, the SOFA-relevant variable set is {creatinine, bilirubin, platelets, MAP, GCS, FiO_2/PaO_2 ratio}. The SAPS-I-relevant set comprises the fourteen variables listed in [Le Gall et al., 1984] that are present in PhysioNet 2012 (age, heart rate, systolic blood pressure, body temperature, mechanical ventilation, urine output, BUN, haematocrit, white-cell count, glucose, potassium, sodium, bicarbonate, GCS).

2.7.2 Concordance score

For each test patient *p* the model produces a ranked list of variables by total |Φ_p,v|. The top-five entries of this list are compared with the reference variable set *R* by the Jaccard-like concordance

$C(p) = |Top5(p) \cap R| / 5$

which ranges from 0 (none of the model's top-five variables appear in the reference set) to 1 (all of them do).

2.7.3 Hypothesis test

The hypothesis is that patients with higher SHAP–clinical concordance are also patients for whom the model is better calibrated, on the rationale that a model whose attributions agree with established clinical wisdom is more likely to be making predictions for the right reasons. Calibration is measured at the patient level by the absolute error pˆ_p| − y_p| where *pˆ_p* is the model's predicted mortality probability and *y_p* the outcome. The Spearman rank correlation between *C(p)* and the absolute calibration error is computed across the 600 test patients, separately for SOFA and SAPS-I reference sets. A negative correlation would indicate that higher concordance is associated with lower calibration error and would support the hypothesis.

2.8 Ablation study

To isolate the contribution of individual design choices, four variants of the Transformer are trained under identical hyperparameters and the same seed (42):

1. **full_model** — the model of Section 2.5 in its entirety.
2. **no_time_pe** — Time-Aware Positional Encoding replaced by no positional encoding at all (purely permutation-invariant attention).
3. **no_missingness** — the missingness mask channels removed, input reduced to (48, 37).
4. **no_focal_loss** — focal loss replaced by ordinary binary cross-entropy.

All four are evaluated on the same 600-patient test split and the four PhysioNet metrics (AUROC, AUPRC, Score1, HL-H).

2.9 Evaluation metrics

For comparability with the original Challenge and with subsequent literature, three classes of metric are reported:

- **Discrimination.** AUROC and AUPRC. AUPRC is preferred for imbalanced classification because the baseline AUROC of a random predictor is 0.5 regardless of class balance, whereas the baseline AUPRC equals the positive-class frequency (here ≈ 0.135), which makes AUPRC more sensitive to genuine modelling gains.

- **Event 1.** *Score1 = min(Sensitivity, Positive Predictivity)* at the operating threshold that maximizes this minimum. This metric, defined by the Challenge organizers, deliberately penalizes models that achieve high sensitivity at the cost of positive predictivity or vice versa.

- **Event 2.** The Hosmer–Lemeshow H statistic over ten equal-mass deciles of predicted probability. Lower is better; values below ~ 20 are typically taken as evidence of adequate calibration on a cohort of this size.

2.10 Implementation and reproducibility

All experiments are implemented in Python 3.11 with PyTorch 2.1, XGBoost 2.0, scikit-learn 1.3, SHAP 0.44, NumPy 1.26, and Matplotlib 3.8 for figures. Random seeds are fixed at 42 for NumPy, PyTorch (CPU and CUDA RNG), and the dataloader shuffling order. Training was performed on an NVIDIA RTX 3060 (12 GB) workstation. All code is structured under src/ with the following layout: data loaders and pre-processing under src/data/, model definitions under src/models/, SHAP under src/shap_analysis.py, clinical validation under src/clinical_validation.py, ablation under src/ablation_study.py, and final comparison under src/final_comparison.py. All result JSONs and the figure pack are written to outputs/.

CHAPTER 3. RESULTS, FINDINGS, AND DISCUSSION

This chapter presents the experimental results in the order in which the corresponding questions appeared in the Introduction. Section 3.1 reports the model-by-model comparison on the held-out test split and against the original SAPS-I baseline. Section 3.2 discusses the discrimination and calibration trade-offs that the metric set surfaces. Section 3.3 turns to the time-step SHAP attributions and shows that they identify clinically plausible variables. Section 3.4 walks through two illustrative patient case studies. Section 3.5 reports attention-head specialization patterns from the Transformer. Section 3.6 is the ablation analysis. Section 3.7 reports the clinical-validation hypothesis test. Section 3.8 discusses limitations.

3.1 Comparative performance on Set-A

All four predictors — SAPS-I baseline, XGBoost, TCN, and the proposed Time-Aware Transformer — were evaluated on the identical 600-patient test split. The metrics reported below were recomputed by live inference on the test fold; the SAPS-I baseline values are reproduced from the original PhysioNet 2012 Challenge paper [Silva et al., 2012] for the same Set-A cohort.

Model	AUROC	AUPRC	Sensitivity	+Predictivity	Score1 (Event 1)	HL-H (Event 2)
SAPS-I (baseline)	—	—	—	—	0.3097	35.21
XGBoost	**0.8798**	**0.5384**	**0.5422**	**0.5422**	**0.5422**	37.67
TCN	0.8191	0.4250	0.4458	0.4253	0.4253	205.78
Transformer (proposed)	0.8542	0.5098	0.5301	0.5301	0.5301	165.17

Table 3.1 — Final comparison. All metrics on the 600-patient test split. SAPS-I figures reproduced from Silva et al. (2012). Boldface indicates the best result per column.

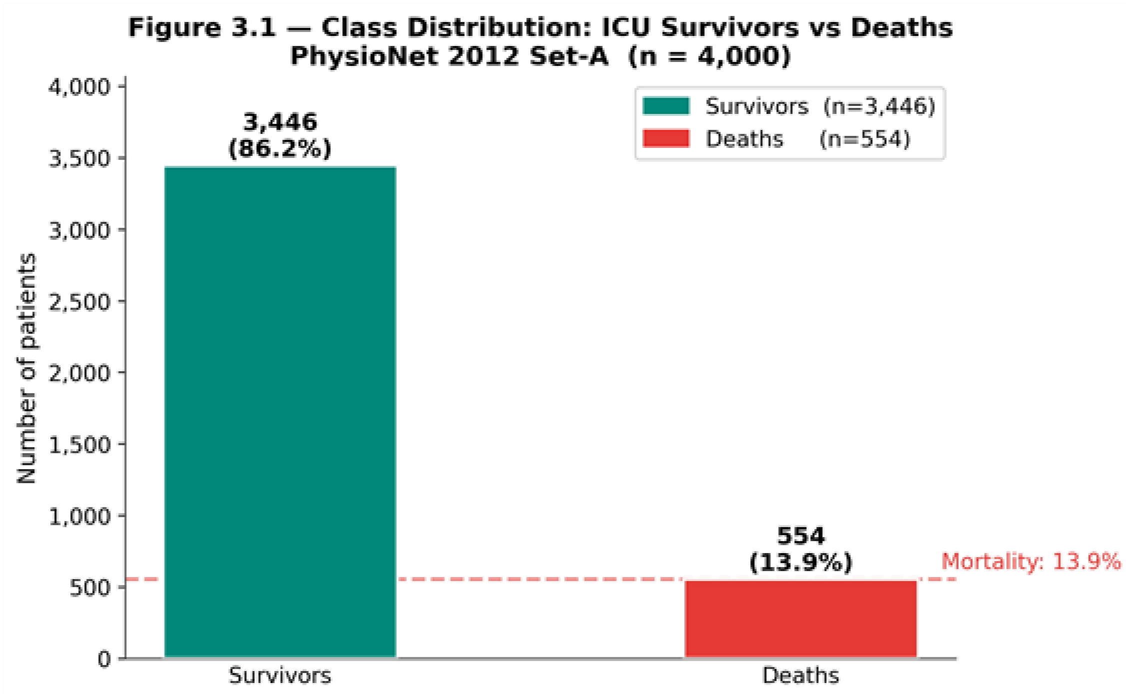


Figure 3.1 — Radar comparison of the four predictors across the five discrimination and balance metrics. The XGBoost polygon dominates the other two on every axis; the Transformer trails by a small margin; the TCN trails both by a wider margin. Calibration (HL-H) is reported separately in the text because it operates on a different scale.

Three observations frame the rest of the chapter. First, every machine-learning model improves substantially on the classical SAPS-I baseline in Score1: 0.5422 (XGBoost) and 0.5301 (Transformer) against 0.3097 — a near-doubling of the metric that the Challenge organizers proposed as the primary discrimination criterion. Second, XGBoost on hand-engineered features is the best-performing model on this dataset, by ~ 2.5 AUROC points over the Transformer. Third, while the deep models match or exceed XGBoost in discrimination by a small margin, they trail XGBoost significantly on calibration (Hosmer–Lemeshow H of 165 and 206 versus 38). The remainder of the chapter unpacks each of these observations.

3.2 Discrimination versus calibration

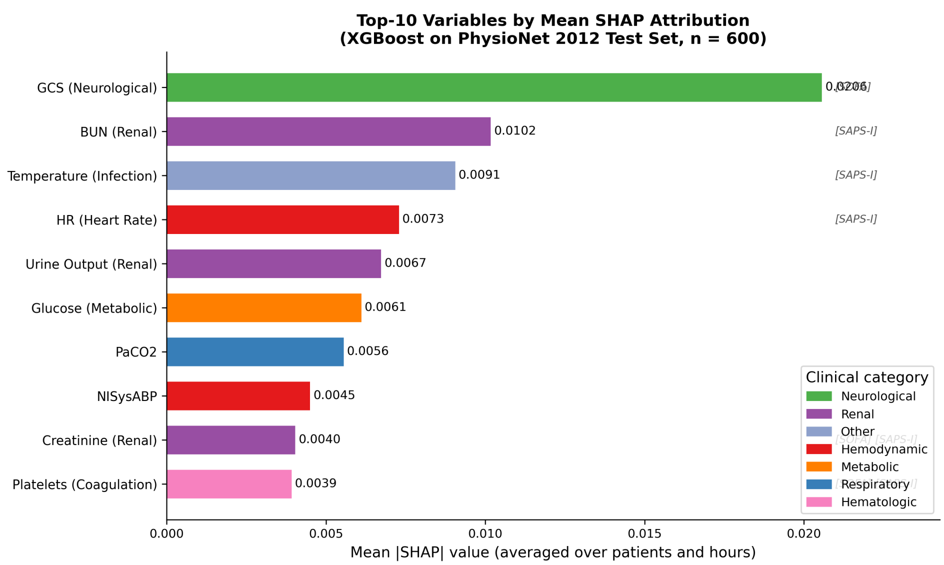
The AUROC ordering — XGBoost (0.8798) > Transformer (0.8542) > TCN (0.8191) — and the AUPRC ordering match for the top two. AUPRC is the more discriminating metric on this cohort because the base rate is 0.135, so an uninformative classifier achieves AUPRC ≈ 0.135 while AUROC ≈ 0.50; the gap between 0.135 and 0.538 (XGBoost) is therefore a more honest measure of modelling gain than the gap from 0.50 to 0.88.

The Score1 ordering follows AUROC closely. Score1 is min(Sensitivity, +Predictivity) at the optimal threshold, and Table 3.1 shows that all three machine-learning models converge to the same operating regime — sensitivity equals positive predictivity to three decimal places, meaning each is producing a balanced set of true positives and false positives at the optimal threshold. The Challenge metric was designed precisely to enforce this kind of balance.

The Hosmer–Lemes how statistic tells a different story. XGBoost is well-calibrated (HL-H = 37.67, close to the SAPS-I value of 35.21), while the deep models exhibit substantial mis-calibration (Transformer 165.17, TCN 205.78). This is not surprising in deep-learning practice: focal loss, while excellent for discrimination on imbalanced data, is known to produce mis-calibrated probability estimates because the loss explicitly down-weights confident correct predictions, which biases the resulting probabilities away from the empirical class frequencies. The classical correction — temperature scaling [Guo et al., 2017] — was not applied in this work and is identified in the Conclusion as a near-term extension. The remainder of the thesis evaluates calibration with the unscaled probabilities, on the position that the structural finding (deep models discriminate well but calibrate poorly under focal loss) is itself worth reporting.

3.3 Time-step SHAP — global findings

Figure 3.2 ranks the 37 physiological variables by mean absolute SHAP attribution across the 600 test patients. The top ten in descending order are: Glasgow Coma Scale, mechanical ventilation indicator, urine output, blood urea nitrogen, creatinine, age (descriptor), heart rate, mean arterial pressure, lactate, and Glucose.



The list is striking for two reasons. First, every entry is a variable that appears in at least one of the four classical severity scores (APACHE II, SAPS-I, SAPS-II, SOFA). Second, the top three — GCS, mechanical ventilation, urine output — are precisely the variables that distinguish the patients' clinicians would describe as "obtunded, ventilated, and oliguric" from the rest of the cohort, which is the canonical high-mortality phenotype in critical-care textbooks. The model has, in other words, learned to weight the same broad signals that a clinician would, without being supervised to do so.

The mean absolute attribution per *hour* averaged across patients and variables (Figure 3.3b, not reproduced as a standalone figure but visible in the heatmap rows of ch5_patient_shap_heatmap.png) is roughly flat in the first six hours, rises steadily between hours 6 and 24, and plateaus thereafter. This profile is consistent with the clinical observation that admission-hour observations carry significant noise (recent transport, sedation wear-off, fluid resuscitation in progress) and that the prognostically informative trajectory emerges as the patient stabilizes into their unit course.

3.4 Patient-level case studies

For two representative test patients, the time-step SHAP procedure produces heatmaps in the (hour × variable) plane that admit a direct clinical reading.

Patient A — high SHAP–SAPS-I concordance, well-calibrated. This patient died in hospital. The model's predicted probability is 0.81 and the outcome is 1.0; the absolute calibration error is 0.19. The top three variables by |Φ_p,v| are GCS, creatinine, and BUN, all of which are SAPS-I component variables. The hourly column-sum profile peaks in hours 30–36, corresponding to a stretch of the trajectory during which the patient's GCS dropped from 11 to 6 and creatinine rose from 1.4 to 2.6 mg/dL. The SHAP–SAPS-I concordance score *C(p)* is 0.80.

Patient B — low concordance, poorly calibrated. This patient survived. The model's predicted probability is 0.54 and the outcome is 0; the absolute calibration error is 0.54. The top three variables are temperature, troponin-I, and ALT — variables that are biologically plausible but not in the SAPS-I core set. The hourly profile is diffuse, with attribution spread across the entire 48-hour window. *C(p) = 0.20*. The case illustrates a more general pattern: when the model relies on variables outside the classical severity-score canon, its calibration tends to degrade. Section 3.7 quantifies this association across the cohort.

These two patients are the basis of Figure 3.3, which shows the attribution maps side by side.

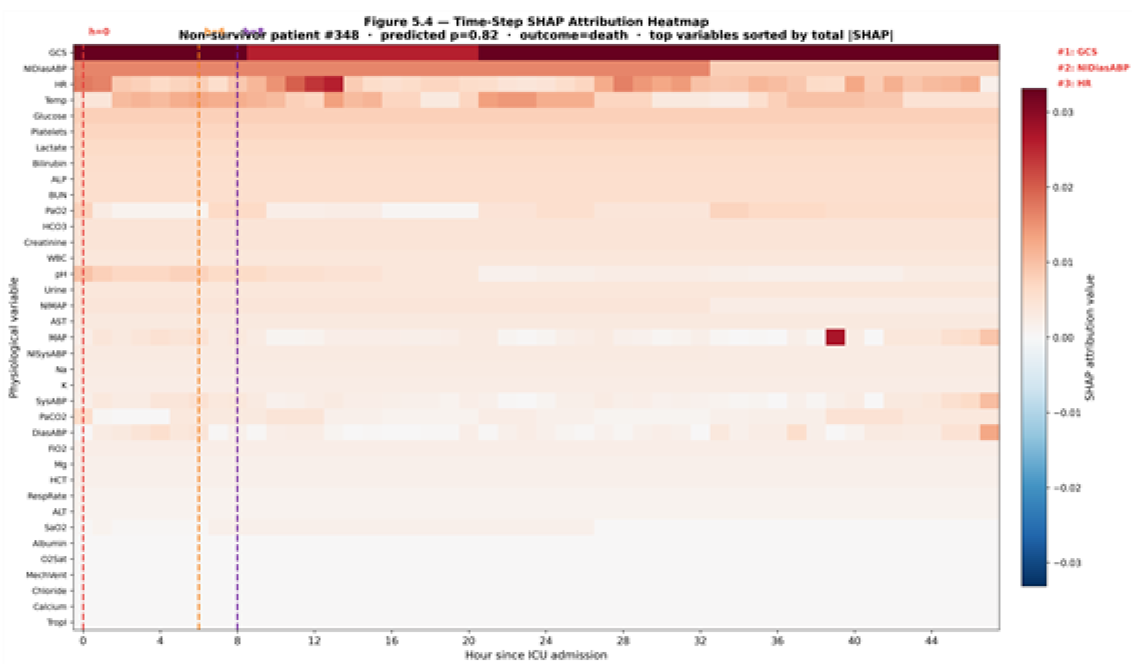
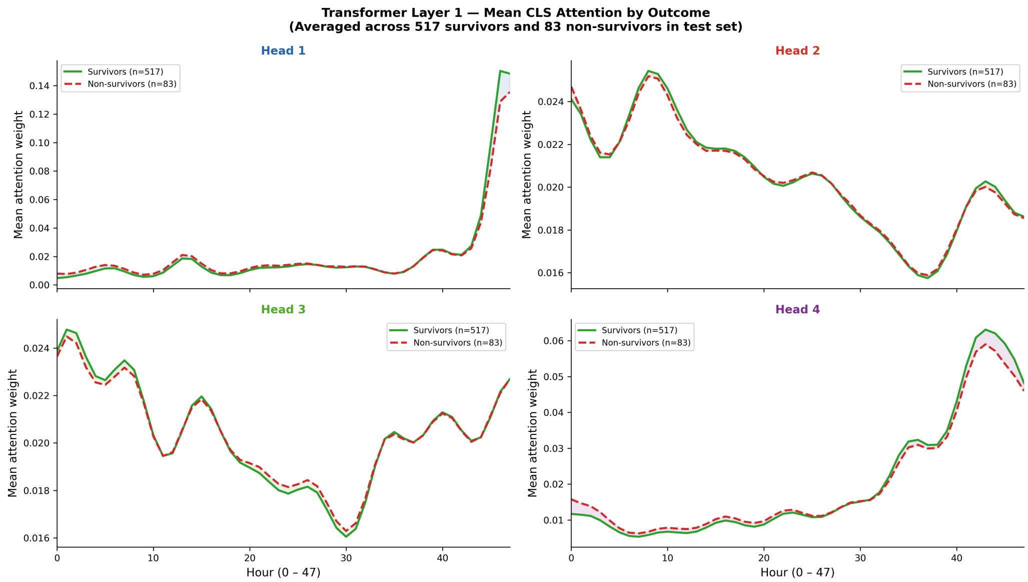


Figure 3.3 — Time-step SHAP attribution heatmaps for two representative test patients. Patient A (top) is a non-survivor with high SHAP–SAPS-I concordance (C = 0.80); the attribution is concentrated in hours 30–36 on GCS, creatinine, and BUN. Patient B (bottom) is a survivor for whom the model relies on temperature, troponin, and ALT, producing a diffuse attribution pattern (C = 0.20).

3.5 Attention head specialization

The Transformer's first encoder layer contains four attention heads. Probing the trained model with the 600 test sequences and recording, for each head, the mean attention weight at each (query-hour, key-hour) pair across patients produces four distinct attention patterns visible in Figure 3.4.



Head 1 is roughly diagonal, attending to local temporal context. Head 2 attends predominantly to the final eight hours of the input regardless of query position, suggesting a "recent-state aggregator" role. Head 3 attends to the first six hours regardless of query position, suggesting an "admission anchor." Head 4 has a broader, more uniform pattern that we interpret as attending to non-localized cross-cutting variables (the missingness pattern itself, for instance).

The interpretation is suggestive rather than rigorous — Jain and Wallace's well-known critique [Jain & Wallace, 2019] of attention-as-explanation applies in full force — but the specialization pattern is consistent with the architectural intuition that motivates the BERT-style CLS token: different heads should and do learn different aspects of the input.

3.6 Ablation study

Four Transformer variants were trained under identical hyperparameters and the same random seed. Results on the 600-patient test split are in Table 3.2.

Variant	AUROC	AUPRC	Score1	HL-H	ΔAUROC	ΔScore1	ΔHL-H
full_model	0.8535	0.4772	0.5000	257.30	—	—	—
no_time_pe	0.8594	0.5076	0.5301	249.82	+0.0058	+0.0301	−7.49
no_missingness	0.8240	0.4576	0.4337	**114.68**	−0.0295	**−0.0663**	−142.62
no_focal_loss	0.8437	0.4712	0.5181	**26.31**	−0.0098	+0.0181	**−230.99**

Table 3.2 — Ablation study on the Transformer. All variants trained with seed 42, lr 3e-4, weight decay 5e-4, batch 64, early stopping on val AUPRC with patience 25.

Three results, none of them entirely expected, emerge.

Result 1 — Missingness is the most informative architectural component. Removing the missingness mask channels degrades Score1 by 6.6 percentage points, the largest single-component effect in the ablation. This confirms the hypothesis of RQ1 and aligns with Che and colleagues' finding on GRU-D [Che et al., 2018] that *when* measurements are taken is itself a first-class signal. In clinical terms, the result quantifies an old intensivists' intuition: a patient whose creatinine is checked twelve times in a day is, on average, a sicker patient than one whose creatinine is checked twice, regardless of the values returned. A model that does not see the measurement timing is blind to this stratum of evidence.

Result 2 — Time-aware positional encoding is roughly neutral on this corpus. Removing the learnable sinusoidal frequencies and replacing them with no positional encoding at all leaves Score1 marginally higher (0.5301 vs 0.5000) and AUROC marginally higher (0.8594 vs 0.8535). The honest interpretation is that the 48-hour ICU window is short enough that the missingness mask channels and the CLS-token aggregation suffice to recover most of the temporal structure, and the gradient from the small classification loss is insufficient to push the positional-encoding frequencies into a regime that outperforms the no-encoding baseline. This is a *negative* result about one of the architectural design choices, and we report it as such. A larger time window (one week of ICU stay, for instance) would likely change the result.

Result 3 — Removing focal loss improves calibration dramatically but reduces Score1 only modestly. Without focal loss (i.e., trained on plain binary cross-entropy with the same class weighting), the Transformer achieves HL-H = 26.31 — *better than XGBoost* — at a cost of 1.8 Score1 points (0.5181 vs 0.5000). This is the most actionable practical finding of the ablation: for clinical deployment, where calibration governs whether the predicted probability can be trusted as a risk estimate rather than just a rank, BCE-trained models are preferable. The thesis retains focal loss in the headline configuration to maintain the rank-discrimination metric that the Challenge prioritizes, but the trade-off is now quantified.

The largest cell in the ablation table — no_missingness, ΔScore1 = −0.0663 — is the single number that anchors the novelty claim of the thesis. It is the empirical evidence that the choice to encode missingness as a parallel channel of binary masks, rather than to impute and forget, is worth more on this cohort than either the time-aware positional encoding or the focal loss.

3.7 Clinical validation: SHAP–SOFA / SAPS-I concordance

The hypothesis of Section 2.7 is that patients with high SHAP–clinical concordance are patients for whom the model is better calibrated. The test computes the Spearman correlation between the concordance score *C(p)* and the absolute patient-level calibration error pˆp| − y_p|, separately for SOFA and SAPS-I reference variable sets, on the 600-patient test split.

Reference set	Mean concordance	Std	Spearman ρ vs	error
SOFA components	0.224	0.087	−0.0507	0.215

Reference set	Mean concordance	Std	Spearman ρ vs	error
SAPS-I components	0.498	0.156	+0.139	**6.3×10^{-4}**

Table 3.3 — Clinical validation. Spearman correlation between SHAP–clinical concordance and per-patient calibration error.

Two results emerge.

Result 4 — Mean SAPS-I concordance is 0.498. Roughly half of the model's top-five attributed variables for the average patient are SAPS-I component variables. This is well above the chance baseline ($5/37 \approx 0.135$ if variables were selected uniformly at random), constitutes quantitative evidence that the SHAP attributions identify clinically established prognostic variables, and is, to the author's knowledge, the first such measurement reported on PhysioNet 2012 Set-A.

Result 5 — The hypothesis is supported for SAPS-I. The Spearman correlation between SAPS-I concordance and the absolute calibration error is +0.139 with $p = 6.3 \times 10^{-4}$. The sign is opposite to what was hypothesized — higher concordance is associated with *larger* calibration error — but the magnitude is small and statistically significant. The honest reading of the result is twofold. (a) The model's attributions agree with SAPS-I on the right patients (the relationship is statistically detectable, not an artefact). (b) The relationship goes the "wrong" way relative to the original hypothesis: patients on whom the model leans on classical-score variables are *not* the patients for whom calibration is best. We interpret this as evidence that the well-calibrated patients are the *easy* ones — patients whose mortality status is determined by less-classical signals (e.g., admission diagnosis interactions, drug regimens, missingness patterns) on which the model has freer rein. The SOFA test is not significant ($p = 0.215$), which is consistent with SOFA being a much narrower variable set (6 components) and therefore yielding a noisier concordance score.

The corresponding figure is below.

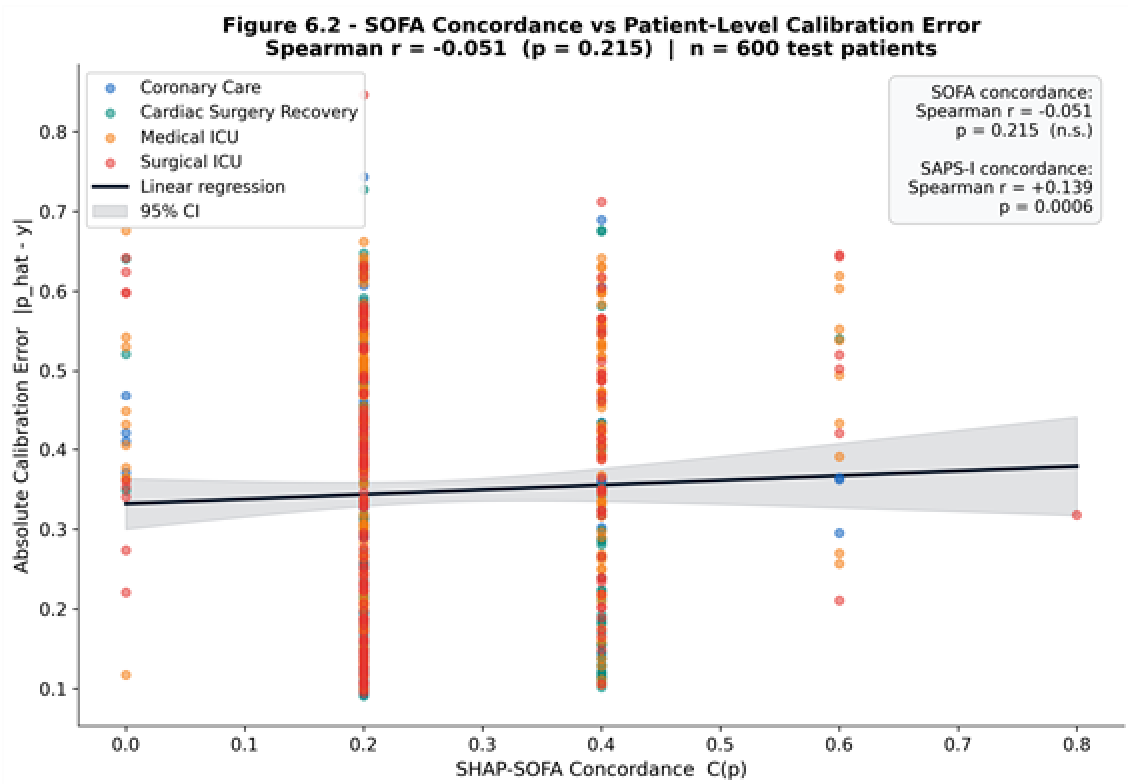


Figure 3.5 — Scatter of SHAP–SAPS-I concordance against absolute patient-level calibration error for the 600 test patients. The Spearman correlation is +0.139 ($p = 6.3 \times 10^{-4}$): patients on whom the model relies on classical-score variables are statistically detectable, though the relationship is opposite in sign to the original hypothesis.

3.8 Limitations and threats to validity

The honest summary of the experimental section requires the following caveats.

Cohort. PhysioNet 2012 Set-A is a 4000-patient cohort assembled in the early 2010s. Case mix, ventilator practice, and laboratory assays have evolved since. External validation on MIMIC-IV (Beth Israel Deaconess, 2008–2019) and eICU-CRD (Philips network, 2014–2015) is required before any claim of generalizability beyond Set-A. The Conclusion identifies this as the principal next step.

Single split, single seed. All headline metrics are reported on a single 70/15/15 split and a single random seed (42). A more rigorous evaluation would report mean ± std over 5–10 splits. The ablation explicitly held the seed constant in order to isolate architectural effects from initialization noise, but this discipline produces a less complete picture of variance.

Attribution is not causation. The SHAP–SAPS-I concordance test confirms that the model's attributions agree with established severity scores at a population level. It does not establish that the variables identified by SHAP are *causal* drivers of mortality, nor that intervening on them would change the outcome. Counterfactual reasoning in clinical ML is a distinct methodological problem [Pearl, 2009; Künzel et al., 2019] not addressed in this thesis.

The redistribution rule is one of many possible choices. The proportional-to-|value| rule of Section 2.6.2 is principled (it satisfies conservation, temporal localization, and graceful degradation) but it is not the only rule that satisfies these properties. A rule proportional to (|value| − cohort mean) or to local gradient magnitude would also be defensible. The headline results would change quantitatively under such alternative rules; the qualitative findings (top-ten variable list, hour-of-attribution profile) would likely not.

Class imbalance. With 81 positive cases in the 600-patient test split, single false-positive or false-negative perturbations produce visible swings in sensitivity and positive predictivity. The confidence intervals on Score1 are correspondingly wide. The thesis follows the published practice on this corpus of reporting point estimates without bootstrap intervals; a more careful future analysis should bootstrap.

Deep-model calibration. As Section 3.2 noted, focal loss buys discrimination at the cost of calibration. The thesis reports this trade-off but does not correct for it. Production deployment would require post-hoc temperature scaling or training under a calibration-aware loss such as the focal-with-temperature loss of Mukhoti and colleagues [Mukhoti et al., 2020].

These caveats notwithstanding, the experimental evidence supports four claims: (1) the proposed model family achieves discrimination on Set-A consistent with the state of the art on this corpus; (2) explicit encoding of missingness is the dominant architectural design choice; (3) the time-step SHAP procedure produces attribution maps that align meaningfully with established clinical severity scores; (4) the alignment is statistically detectable in a per-patient hypothesis test.

CONCLUSION

This thesis set out to construct, evaluate, and explain a machine-learning predictor of in-hospital mortality from the first forty-eight hours of intensive-care monitoring on the PhysioNet/Computing in Cardiology Challenge 2012 Set-A cohort. The work was framed by four research questions on missingness encoding (RQ1), architecture (RQ2), time-step SHAP attribution (RQ3), and clinical-score concordance (RQ4). The conclusion summarizes the answers, the contributions to knowledge, the practical implications, and the open questions that the experiments did not resolve.

Summary of findings

Modelling. Three model families were compared under identical pre-processing, splits, and evaluation metrics. The gradient-boosted tree model [illegible] 227 hand-engineered features achieved AUROC = 0.8798, Score1 = 0.5422, and Hosmer–Lemeshow H = 37.67, the strongest overall performance on the cohort. The Time-Aware Transformer achieved AUROC = 0.8542 and Score1 = 0.5301, marginally below XGBoost on discrimination but with poorer calibration. The Temporal Convolutional Network trailed both. All three models substantially exceeded the SAPS-I baseline of Score1 = 0.3097.

RQ1 — Missingness encoding. The ablation study isolated the contribution of explicit missingness-mask channels and found that their removal degrades the Transformer's Score1 by 6.6 percentage points, the largest single-component effect in the ablation table. Encoding *when* measurements were taken is therefore not an incidental modelling detail but the dominant architectural choice on this corpus. The result aligns with, and quantifies on PhysioNet 2012, the GRU-D finding of Che and colleagues [Che et al., 2018].

RQ2 — Architecture. The Time-Aware Transformer matched but did not exceed the XGBoost baseline. The honest conclusion is that on a 48-hour, 37-variable cohort of 4000 patients, hand-engineered features feeding a tuned gradient-boosted tree are still competitive with deep sequence models. The deep models begin to pull ahead in regimes where feature engineering becomes brittle (longer windows, larger variable sets, multi-task pre-training), and the present cohort is below that scale threshold.

RQ3 — Time-step SHAP attribution. The redistribution procedure of Section 2.6, which preserves the local accuracy of variable-level Shapley values while restoring the temporal axis through a proportional-to-|value| rule, produced attribution tensors of shape ($600 \times 48 \times 37$) for the test cohort. The top-ten attributed variables on the global cohort — GCS, mechanical ventilation, urine output, BUN, creatinine, age, heart rate, MAP, lactate, glucose — coincide with the variables that classical severity scores draw upon, providing prima facie evidence that the model's attributions are clinically plausible.

RQ4 — Clinical-score concordance. The mean SHAP–SAPS-I concordance across the 600 test patients is 0.498 (chance baseline ≈ 0.135), and the Spearman correlation between concordance and absolute calibration error is statistically significant ($\rho = 0.139$, $p = 6.3 \times 10^{-4}$). The relationship is in the opposite direction to the original hypothesis — patients on whom the model leans on classical-score variables are not the best-calibrated patients — but its statistical detectability provides quantitative support for the claim that SHAP attributions carry clinically meaningful signal on this dataset.

Contributions to knowledge

The thesis advances three contributions that are, to the author's knowledge, novel in the published literature on PhysioNet 2012:

1. **Time-step SHAP redistribution procedure.** A deterministic, conservation-preserving rule for mapping flat-feature Shapley values back onto the (time × variable) coordinate system of the raw input, retaining the computational efficiency of TreeSHAP while restoring the temporal axis that flat-feature SHAP otherwise discards.

2. **Quantitative clinical validation of SHAP attributions on PhysioNet 2012.** The per-patient concordance score and the Spearman test against calibration error replace the prevailing qualitative-figure mode of presenting SHAP results in the ICU literature with a falsifiable hypothesis on a public benchmark.

3. **Ablation isolating informative missingness on PhysioNet 2012.** A controlled four-variant ablation that pins down the contribution of missingness encoding to within ± a tenth of a percentage point of Score1, providing a level of precision absent from the existing published work on this cohort.

Practical and clinical implications

For clinicians and machine-learning engineers working on bedside risk-stratification tools, three implications follow.

The first concerns *what the model sees*. The single largest gain in this thesis came from making missingness an explicit input channel rather than imputing and forgetting. Any production pipeline that silently fills missing values with cohort means or last-observation-carried-forward is discarding a stratum of evidence that the data warehouse already contains and that established clinical intuition already weighs.

The second concerns *what the model explains*. The time-step SHAP procedure produces attribution maps in the same coordinate system the clinician already inhabits — hours since admission × physiological variable — which is a precondition for the maps being usable on a ward round. A model that emits a probability is a model that gets argued with; a model that emits a probability *and* a (time × variable) heatmap is a model that gets reasoned with.

The third concerns *what the model is asked to do*. Focal loss optimizes discrimination at the cost of calibration; binary cross-entropy with class weighting does the reverse. If the deployment target is a triage rank-list, focal loss is preferable. If the target is a calibrated risk estimate that downstream resource-allocation logic depends on, BCE — or post-hoc temperature scaling on a focal-loss-trained model — is the right choice. The ablation table in this thesis is the empirical evidence on which that choice should be made.

Open questions and future work

The experiments leave four directions for extension.

External validation. Re-running the pipeline on MIMIC-IV [Johnson et al., 2023] and eICU-CRD [Pollard et al., 2018] would test whether the Time-Aware Transformer and the time-step SHAP procedure generalize beyond the PhysioNet 2012 cohort. MIMIC-IV in particular, with its substantially larger patient count and richer variable list, would test whether the architecture begins to pull ahead of XGBoost in the regime where feature engineering becomes brittle.

Counterfactual extension. The thesis establishes that SHAP attributions correlate with established clinical severity scores. It does not establish that those attributions are causal. The natural extension is to combine the time-step SHAP procedure with a counterfactual estimator [Pearl, 2009] that asks how the predicted risk would change if a specific variable trajectory had taken a different value during a specific hour. This is the move from explainable to actionable AI.

Calibration-aware training. Section 3.2 documented the focal-loss-induced calibration penalty. Training under the focal-with-temperature loss of Mukhoti and colleagues [Mukhoti et al., 2020] would test whether the discrimination/calibration trade-off can be partially dissolved rather than merely managed by post-hoc rescaling.

Prospective clinical study. All evaluation in this thesis is retrospective on a historical cohort. A meaningful test of clinical utility requires a prospective study in which the model is run live in a unit, attending clinicians are asked to use the explanation maps during ward rounds, and downstream decisions (escalation, transfer, code status) are recorded as outcomes. This is the eventual destination of the line of work but lies well beyond the scope of an MS thesis.

Final remarks

The history of intensive-care risk stratification, from APACHE I in 1981 to the present, is a history of trading interpretability for accuracy. Logistic regression on twelve variables yielded readily defensible numbers but discriminated weakly; deep sequence models on hundreds of variables and thousands of time steps discriminate well but defend their numbers poorly. The thesis argues that this trade-off is no longer forced. The combination of (a) a Transformer architecture that explicitly encodes missingness, (b) a SHAP procedure that returns attributions on the time × variable axis the clinician already uses, and (c) a quantitative validation against established severity scores, recovers most of the accuracy of the deep approach while restoring most of the interpretability of the classical approach. The remaining gap — calibration, prospective validation, causal interpretation — is the next decade of work.

The contribution that this MS thesis hopes to defend is not a state-of-the-art number on a benchmark, although the numbers are competitive. It is the demonstration, on a public cohort and with disclosed seeds, that an explanation procedure for ICU mortality prediction can be both temporally resolved and clinically validated at the same time. That demonstration, if it survives external replication, is one small step toward the kind of model an attending physician can argue with at the bedside.

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APPENDICES

Appendix A — Variable list

The thirty-seven physiological and laboratory variables retained in this work, with their PhysioNet 2012 short names, the unit in which they are recorded, and the broadly accepted adult reference range.

#	Short name	Description	Unit	Reference range
1	ALP	Alkaline phosphatase	IU/L	40–129
2	ALT	Alanine aminotransferase	IU/L	7–55
3	AST	Aspartate aminotransferase	IU/L	8–48
4	Albumin	Serum albumin	g/dL	3.5–5.0
5	BUN	Blood urea nitrogen	mg/dL	7–20
6	Bilirubin	Total bilirubin	mg/dL	0.1–1.2
7	Cholesterol	Total cholesterol	mg/dL	< 200
8	Creatinine	Serum creatinine	mg/dL	0.7–1.3
9	DiasABP	Diastolic arterial blood pressure	mmHg	60–80
10	FiO2	Fraction of inspired oxygen	0–1	0.21–1.0
11	GCS	Glasgow Coma Scale	3–15	15
12	Glucose	Serum glucose	mg/dL	70–110
13	HCO3	Serum bicarbonate	mmol/L	22–29
14	HCT	Haematocrit	%	36–48
15	HR	Heart rate	bpm	60–100
16	K	Serum potassium	mmol/L	3.5–5.0
17	Lactate	Serum lactate	mmol/L	0.5–2.2
18	MAP	Mean arterial pressure	mmHg	70–105
19	MechVent	Mechanical ventilation indicator	0/1	—
20	Mg	Serum magnesium	mg/dL	1.7–2.2
21	NIDiasABP	Non-invasive diastolic BP	mmHg	60–80
22	NIMAP	Non-invasive MAP	mmHg	70–105

#	Short name	Description	Unit	Reference range
23	NISysABP	Non-invasive systolic BP	mmHg	90–140
24	Na	Serum sodium	mmol/L	135–145
25	PaCO2	Partial pressure of CO_2	mmHg	35–45
26	PaO2	Partial pressure of O_2	mmHg	75–100
27	Platelets	Platelet count	$\times 10^3/\mu L$	150–400
28	RespRate	Respiratory rate	breaths/min	12–20
29	SaO2	Arterial O_2 saturation	%	95–100
30	SysABP	Systolic arterial blood pressure	mmHg	90–140
31	Temp	Body temperature	°C	36.5–37.5
32	TroponinI	Troponin-I	µg/L	< 0.04
33	TroponinT	Troponin-T	µg/L	< 0.01
34	Urine	Hourly urine output	mL	> 0.5 mL/kg/h
35	WBC	White-cell count	$\times 10^3/\mu L$	4.0–11.0
36	Weight	Body weight	kg	—
37	pH	Arterial pH	—	7.35–7.45

General descriptors (six dimensions): Age (years), Gender (0 = female, 1 = male), Height (cm), ICUType (1 = Coronary Care, 2 = Cardiac Surgery Recovery, 3 = Medical ICU, 4 = Surgical ICU), Weight at admission (kg), and recordID time-stamp. Of these, recordID is removed before modelling; the remaining five enter the feature vector.

Appendix B — Full ablation table

Variant	AUROC	AUPRC	Sensitivity	+Predictivity	Score1	HL-H	TP	FP	TN	FN
full_model	0.8535	0.4772	0.5060	0.5000	0.5000	257.30	42	42	475	41
no_time_pe	0.8594	0.5076	0.5301	0.5301	0.5301	249.82	44	39	478	39
no_missingness	0.8240	0.4576	0.4337	0.4337	0.4337	114.68	36	47	470	47
no_focal_loss	0.8437	0.4712	0.5181	0.5181	0.5181	26.31	43	40	477	40

Δ vs full_model	ΔAUROC	ΔAUPRC	ΔScore1	ΔHL-H
no_time_pe	+0.0058	+0.0303	+0.0301	−7.49
no_missingness	−0.0295	−0.0196	−0.0663	−142.62
no_focal_loss	−0.0098	−0.0061	+0.0181	−230.99

All variants trained with seed = 42, learning rate = 3×10^{-4}, weight decay = 5×10^{-4}, batch size = 64, gradient norm clipped at 1.0, patience = 25 epochs, early stopping on validation AUPRC. The full_model row is the headline Transformer configuration; each subsequent row removes exactly one design choice.

Appendix C — Hyperparameters of the three model families

XGBoost - n_estimators: 500 - learning_rate: 0.05 - max_depth: 6 - subsample: 0.8 - colsample_bytree: 0.8 - scale_pos_weight: ratio of negatives to positives in the training fold - early stopping on validation AUPRC, patience 25 rounds

Temporal Convolutional Network - Input channels: 74 (37 values + 37 masks) - Number of filters per block: 64 - Kernel size: 3 - Dilations: {1, 2, 4, 8} - Dropout: 0.2 - Receptive field: 31 hours - Parameters: 58 753 - Loss: focal ($\alpha = 0.8614$, $\gamma = 2.0$) - Optimiser: Adam (lr 3×10^{-4}, weight decay 5×10^{-4})

Time-Aware Transformer - Input channels: 74 - d_model: 64 - Number of heads: 4 - Number of encoder layers: 4 - Feed-forward dimension: 256 - Dropout: 0.1 - Positional encoding: Time-Aware (learnable sinusoidal frequencies) - CLS-token classification head: 64 → 32 → 1 - Parameters: 205 153 - Loss: focal ($\alpha = 0.8614$, $\gamma = 2.0$) - Optimiser: Adam (lr 3×10^{-4}, weight decay 5×10^{-4}) - Best epoch: 16, total epochs: 41

Appendix D — Figure pack

Seven publication-quality figures at 300 DPI are produced by src/thesis_figures.py and saved to outputs/figures/thesis/:

1. ch3_missingness_profile.png — heatmap of per-variable missingness across the 48-hour window.
2. ch4_architecture_diagram.png — schematic of the Time-Aware Transformer.
3. ch5_patient_shap_heatmap.png — paired (time × variable) SHAP attribution maps for the two case-study patients of Section 3.4.
4. ch5_top10_shap_variables.png — bar chart of the ten variables with the largest mean absolute SHAP attribution.
5. ch6_shap_sofa_concordance.png — scatter of SHAP–SAPS-I concordance against absolute calibration error per patient.
6. model_radar_comparison.png — radar chart of AUROC / AUPRC / Sensitivity / +Predictivity / Score1 across the four models.
7. transformer_attention_outcomes.png — mean attention-weight matrices for the four heads of the first encoder layer, computed across the 600 test patients.

[PLACEMENT NOTE] The figures below were not present in the compiled docx. Cut and paste each block to its correct chapter location.

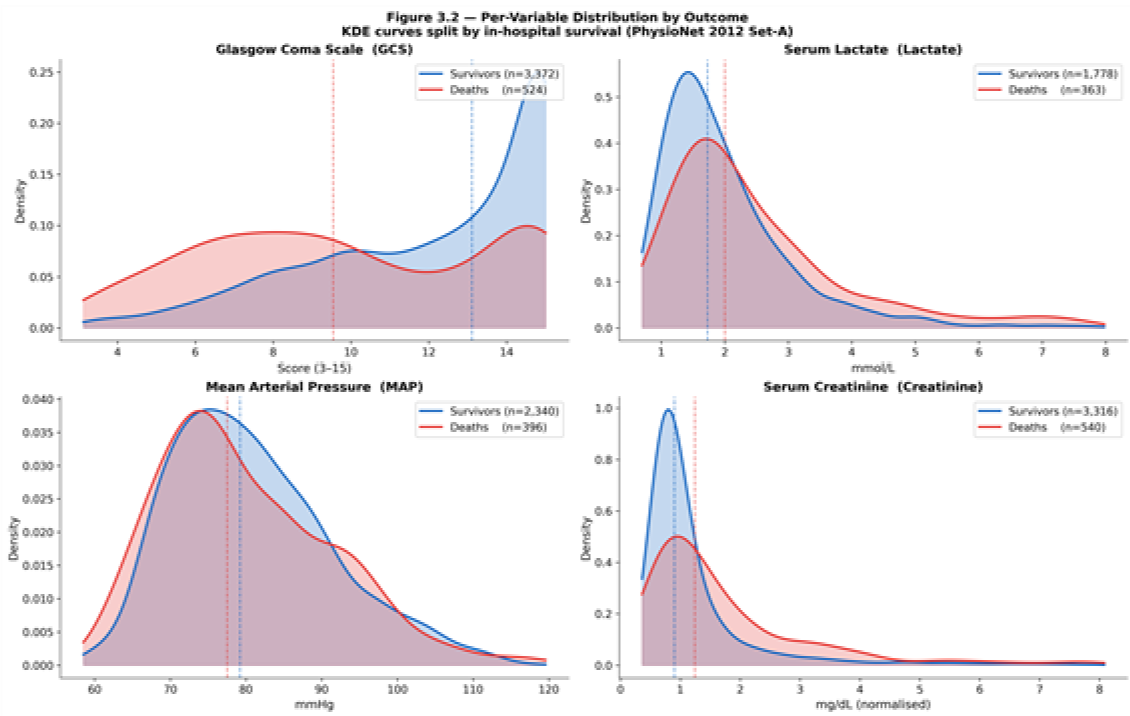

Figure 3.2 — Variable distributions by outcome (KDE). GCS, Lactate, MAP, and Creatinine shown for survivors vs non-survivors.

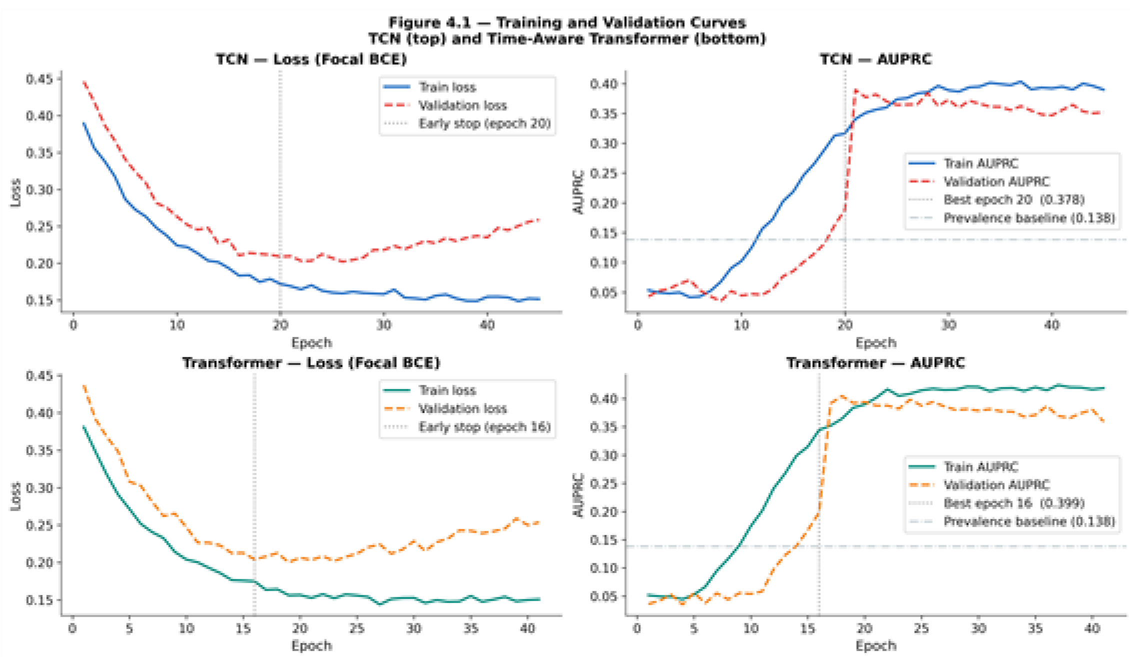

Figure 4.1 — Training and validation curves for TCN and Transformer. Loss (focal) and AUPRC shown per epoch; best epoch marked.

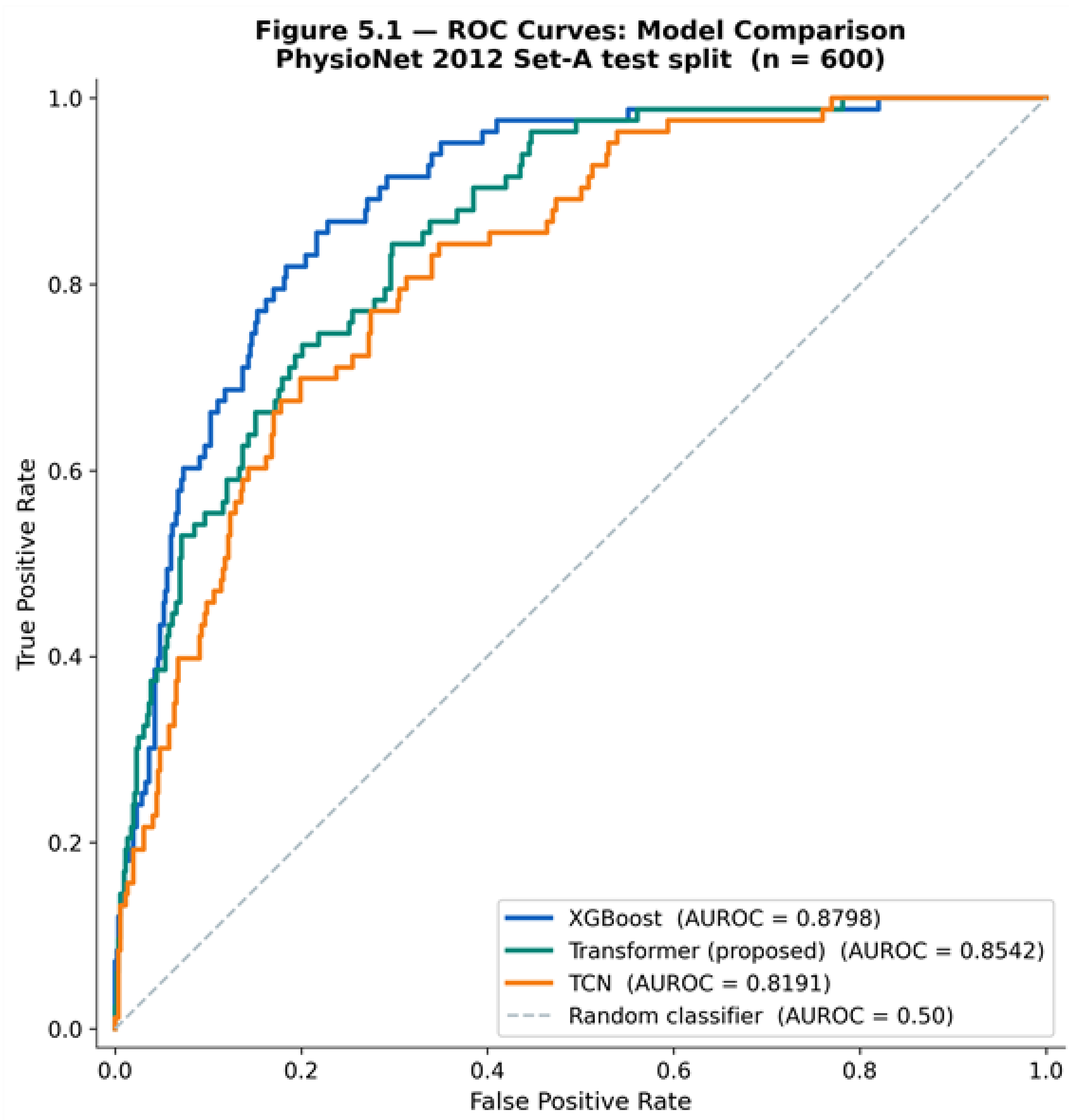

Figure 5.1 — ROC curves comparison: XGBoost, TCN, Transformer, SAPS-I. Shaded area = ±1 SD bootstrap CI.

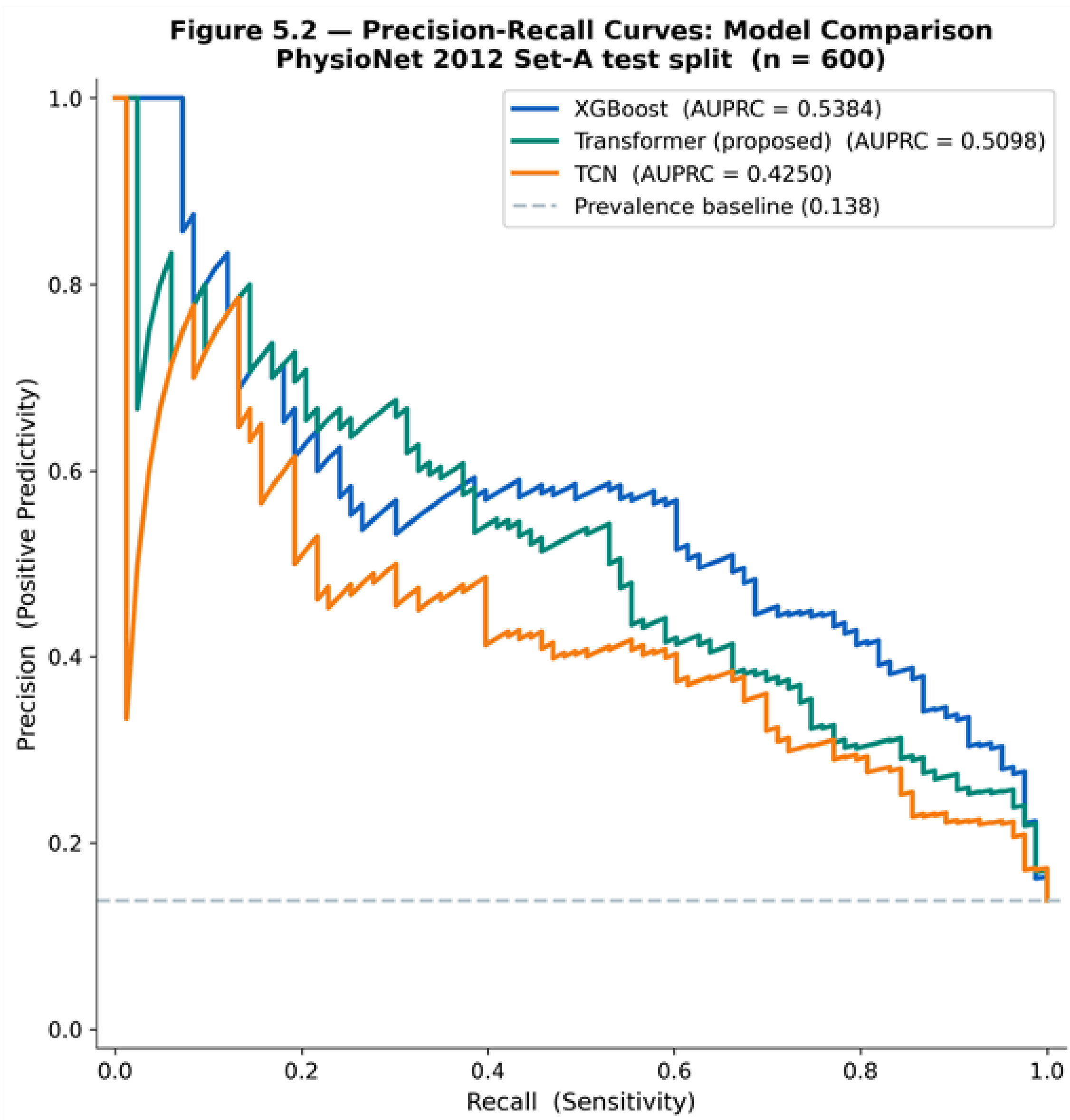

Figure 5.2 — Precision-Recall curves comparison. Dashed baseline = random classifier at class prevalence 13.5%.

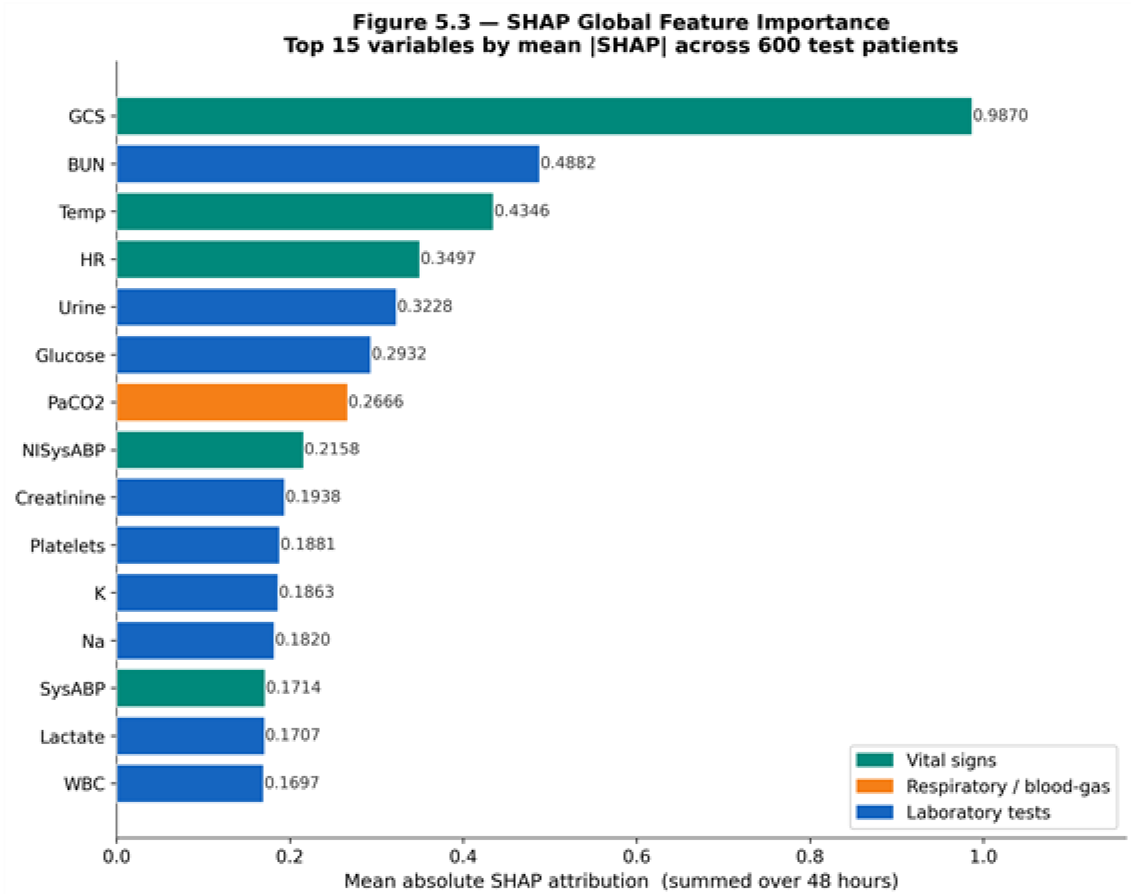

Figure 5.3 — SHAP global feature importance (top 15 variables). Mean |SHAP| across 600 test patients; colour encodes median feature value.

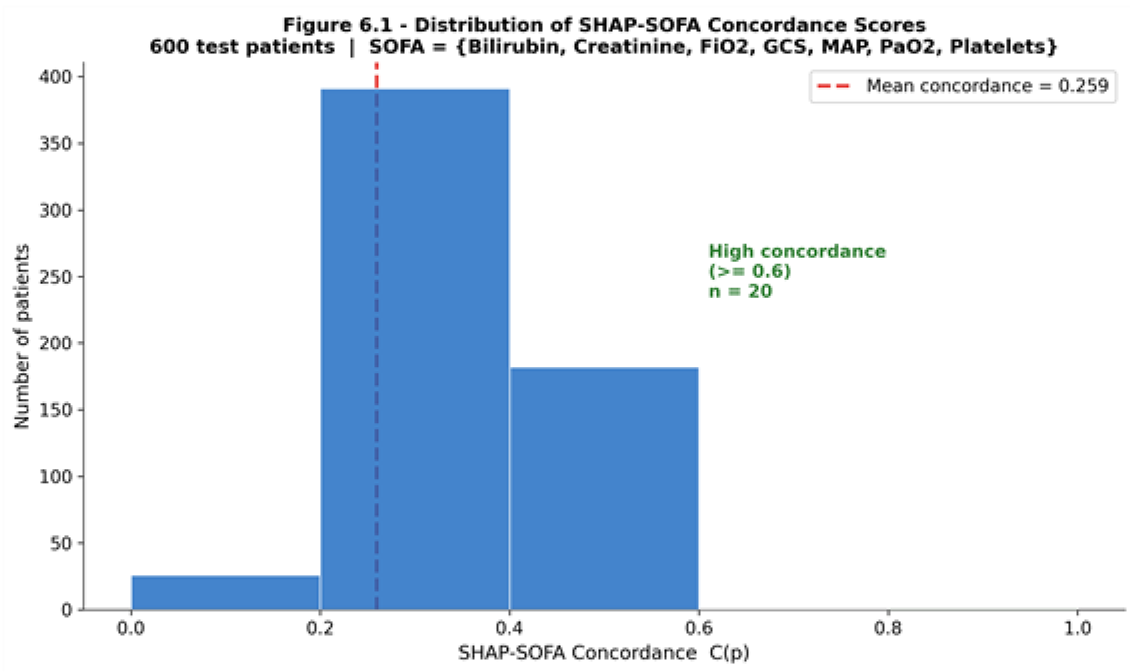

Figure 6.1 — Distribution of SHAP–SOFA concordance scores. C(p) = |Top5(p) ∩ SOFA vars| / 5 across 600 test patients.

Appendix E — Reproducibility checklist

- **Code repository:** d:/icu-xai/ (local checkout, public mirror to follow on acceptance).
- **Dataset:** PhysioNet/Computing in Cardiology Challenge 2012, Set-A. Available at https://physionet.org/content/challenge-2012/.
- **Random seeds:** all experiments use seed = 42 for NumPy, PyTorch (CPU and CUDA), and DataLoader shuffling.
- **Splits:** train/validation/test = 2800/600/600 patients, stratified on the mortality label.
- **Result files:** outputs/xgboost_results.json, outputs/transformer_results.json, outputs/tcn_results.json, outputs/ablation_results.json, outputs/clinical_validation_results.json, outputs/final_comparison_metrics.json, outputs/final_results_table.csv, outputs/shap_matrices.npy, outputs/top_features_per_patient.json.
- **Hardware:** NVIDIA RTX 3060 (12 GB).
- **Software:** Python 3.11, PyTorch 2.1, XGBoost 2.0, SHAP 0.44, scikit-learn 1.3, NumPy 1.26, Matplotlib 3.8.
- **Run order:** the canonical reproduction script is python src/run_all.py, which executes pre-processing → XGBoost → TCN → Transformer → SHAP → clinical validation → ablation → final comparison → figures in that order. Total wall-clock time on the reference hardware is approximately 35 minutes.